



Review Article

Bitcoin reimaged: A comprehensive study of ordinals and inscriptions protocols for Web3 asset innovation

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ARTICLE INFO

Keywords:

Bitcoin
Blockchain
BRC-20
DeFi
Inscriptions
Ordinals

ABSTRACT

The unprecedented rise of Bitcoin has marked a significant milestone in the evolution of decentralized finance (DeFi). Despite Bitcoin's groundbreaking contributions, it faces inherent challenges due to its reliance on the Unspent Transaction Output model, which limits its capabilities in executing complex transactions and embedding diverse data types. To overcome these limitations, Ordinals and Inscriptions have been introduced, allowing extensive data and information embedding within Bitcoin transactions. Building upon these advancements, the recent development of the Bitcoin Request for Comment (BRC)-20 standard leverages these capabilities to introduce fungible tokens on the Bitcoin network. However, these innovations have yet to be fully integrated into the broader blockchain and DeFi ecosystems, and they still face challenges related to interoperability, scalability, and user perception. In this paper, we explore the intricacies of Ordinals, Inscriptions, and BRC protocols to address these challenges by analyzing their functionalities, operational methodologies, and potential applications. We offer a detailed examination of the challenges and future prospects, shedding light on the unexplored potential of these technologies in transforming Bitcoin transactions and expanding its role in the DeFi space. By thoroughly analyzing these new developments, we aim to bridge the gap in current academic research and offer valuable insights for developers, investors, and enthusiasts. This paper serves as a foundation for future innovations, paving the way for more robust, scalable, and user-friendly applications in the DeFi and Web3 landscape.

1. Introduction

Bitcoin [1] has emerged as the pioneering cryptocurrency, establishing itself as a dominant force within the digital asset landscape. As of now, Bitcoin is valued at 590.74 billion USD [2], making it the 10th largest asset globally and dominating over 53% of the market share within the cryptocurrency space. Following Bitcoin, Ethereum [3], the second-largest cryptocurrency, offers a versatile platform for decentralized applications and smart contracts. Ethereum holds a significant market share of 19.1% [4], though still trailing behind Bitcoin's dominance. Bitcoin has played a crucial role in transforming the digital asset landscape, laying the groundwork for the development of decentralized finance (DeFi) [5]. While Ethereum has been instrumental in the proliferation of DeFi applications, Bitcoin's foundational principles of

decentralization and security have inspired the broader adoption of decentralized financial systems. However, the Unspent Transaction Output (UTXO) model of Bitcoin introduces certain limitations, particularly in handling complex transactions and embedding data, both of which are increasingly crucial in today's digital transaction environment.

To address these constraints, Bitcoin has embraced new technologies known as *Ordinals* and *Inscriptions*, which allow for the embedding of extensive data within transactions [6]. Ordinals assign a unique identity to each satoshi, the smallest unit of Bitcoin, while Inscriptions enable the integration of diverse data types, from text to complex images, directly onto the blockchain. This evolution not only expands Bitcoin's utility but also opens up avenues previously limited to Ethereum and other blockchain ecosystems. The emergence of Ordinals and Inscriptions on Bitcoin presents a potential challenge to Ethereum's established

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Received 25 June 2024; Received in revised form 24 May 2025; Accepted 3 August 2025

dominance in the Non-Fungible Token (NFT) market. Traditionally, Ethereum's smart contract functionality has facilitated a thriving NFT ecosystem, where each NFT represents verifiable ownership of digital assets stored off-chain [7]. However, the integration of Ordinals and Inscriptions opens doors for the creation and trading of Bitcoin-based NFTs, signifying a potential shift in the landscape of digital asset ownership. This development is evidenced by the increasing number of NFT transactions on platforms like Magic Eden [8], which have broadened their scope to encompass Bitcoin NFTs, directly competing with Ethereum-centric marketplaces. While the long-term impact on market share remains to be seen, the rise of Bitcoin-based NFTs signifies a potential shift in the landscape of digital asset ownership. The Bitcoin Request for Comment (BRC) token standard emerges from Bitcoin's new Ordinal and Inscription capabilities [9]. For example, BRC-20, developed by the pseudonymous developer named Domo [10], mirrors the functionalities of Ethereum's ERC-20 but without reliance on smart contracts. This innovation enables Bitcoin not only to host data within its blockchain but also to manage digital assets through BRC-20 tokens, thereby enhancing its role within the digital asset space. This token framework encapsulates a broad spectrum of metadata directly on the Bitcoin network, allowing for the creation of NFTs and expanding Bitcoin's utility in digital asset management. Therefore, the integration of Ordinals, Inscriptions, and BRC tokens marks a significant milestone in Bitcoin's evolution. This development has the potential to reshape its role within the digital asset space by enabling the creation and management of NFTs, enriching Bitcoin's transactional capabilities, and aligning it more closely with the broader functionalities of blockchain, DeFi [11,12], and Web3 [13–15] ecosystems.

1.1. Motivation of this study

While Ordinals, Inscriptions, and BRC protocols have significantly impacted the Bitcoin and DeFi landscapes, these advancements have yet to be extensively explored within academic research regarding their functionalities and potential applications. Recent studies [16–18] have focused specifically on the BRC-20 standard in terms of its influence on user perception and interoperability with Ethereum's ERC-20. For example, Wang and Yu [16] present a comprehensive analysis of the novel BRC-20 token standard in Bitcoin, differentiating it from Ethereum's ERC-20 tokens. The authors employ a comprehensive approach by exploring the open-access resources and conducting extensive quantitative and qualitative analyses. This includes investigating market dynamics and user sentiments for selected BRC-20 and ERC-20 tokens over four months, analysis of token price, market capitalization, user responses on social media, and historical token prices on the new market trends. The study's significant contribution includes the explanation of BRC-20's operational mechanisms and a comparative market study to highlight its potential and acknowledge its current limitations against the established ERC protocol. On the other hand, Yu et al. [17] present a middleware solution for bridging the gap between the Bitcoin and Ethereum networks by leveraging the BRC-20 standard to interpret Bitcoin Inscription and initiate corresponding actions on Ethereum. The authors address the critical challenge of enabling interaction between these distinct blockchain architectures to highlight the flexibility and potential applications of Bitcoin's BRC-20 standard in broader contexts. The research contributes a prototype showcasing functionality between the two networks and evaluates scalability, performance, and operational efficiency on the Ethereum testnet [19]. However, the approach is limited to one-way interactions from Bitcoin to Ethereum, lacks a comprehensive security analysis, and does not explore other BRC protocols. Bertucci [18] presents an insightful study on Bitcoin Ordinal, focusing on the economic analysis of these on-chain non-fungible assets on the Bitcoin from December 14th, 2022, to April 16th, 2023. The author emphasizes statistics, cross-sectional, and regression analyses to provide a nuanced understanding of Bitcoin Ordinal and their effect on transaction fees and miners' revenue. However, the research does not consider

the demand and supply factors affecting transaction fees, the externalities on other Bitcoin applications, and lacking a comparative analysis with other blockchain-based NFTs. Despite valuable insights from existing research, there is still a gap in understanding how these novel protocols can expand Bitcoin's utility beyond its core function as a peer-to-peer (P2P) payment system. This expansion could potentially attract new users and foster innovative applications within the DeFi, NFT, and Web3 landscapes.

1.2. Comparison with existing studies

A recent surge in research activity has illuminated both the substantial progress and the persistent challenges within the blockchain and DeFi ecosystems. For example, Razi et al. [20] provided a foundational understanding of NFT technology to explore its core components, applications, and future challenges by emphasizing the role of NFTs in digital asset management and decentralized identity. The authors discuss a wide range of token standards and NFT applications across various domains, including healthcare, supply chain, gaming, identity verification, agriculture, intellectual property, smart cities, charity, donation, and education. Similarly, Kraussl and Tugnetti [21] explored the NFT market in order to concentrate on factors that determine pricing, investor perceptions, and how these elements influence NFT behavior. They present a framework to understand the price formation of NFTs and its influence on market dynamics. Likewise, Schwiderowski et al. [22] classified crypto token types into asset tokens, utility tokens, and payment tokens and provide a structured understanding of the various dimensions and characteristics of these systems on the blockchain economy. Meanwhile, Hammi et al. [23] deep dived into the technical features and applications of NFTs, particularly their potential to enhance the metaverse and decentralized identity management. On the other hand, Wang and Nixon [24] offered a systematic study of tokenization on blockchain and classify tokens into fungible, non-fungible, and semi-fungible categories. Their work focuses into the critical challenges and future research directions to make tokenization more efficient and scalable. In terms of metaverse and NFTs, Belk et al. [25] presented new forms of ownership, such as fractional ownership to explore consumer motivations and the implications of these digital assets on traditional concepts of ownership. Environmental concerns surrounding cryptocurrency mining are addressed by Heinonen et al. [26] as they proposed various technological solutions, including hybrid proof-of-stake, sustainable energy sources, and unconventional computing methods, to make cryptocurrencies greener and more energy-efficient. Fang et al. [27] provided a comprehensive survey of cryptocurrency trading research to analyze trading systems, strategies, and risk management by identifying trends and opportunities for future research in this rapidly evolving blockchain ecosystem.

Our study distinguishes itself from existing surveys in several key aspects. First, we adopt a more action-oriented approach, focusing on the potential applications of Ordinals, Inscriptions, and BRC protocols within DeFi and Web3. While prior works [20,21] provide valuable insights into NFTs and their pricing structures, their focus lies primarily on the current state and challenges of the technology. These studies offer limited exploration of how NFTs can be leveraged for future applications. In contrast, our research investigates the technical functionalities of Ordinals, Inscriptions, and BRC protocols. We analyze how these innovations enable the creation and manipulation of data directly on the Bitcoin blockchain. This capability paves the way for novel DeFi and Web3 use cases beyond the simple representation of assets. Additionally, existing token-based studies [22,24] primarily focus on categorizing different types of tokens. However, they fall short in providing an in-depth analysis of the underlying technical mechanisms that enable advanced functionalities. Our research addresses this gap by thoroughly examining the integration of BRC protocols with Bitcoin's UTXO model. This integration is crucial as it facilitates secure token deployment, transfer, and management, thereby enhancing the overall functionality and

Table 1
Comparison of existing surveys.

Literature	Token classification	Technical representation	Scalability and interoperability	DeFi/Web3 applicability	Challenges and limitations	Future direction
Razi et al. [20]	✓	✗	✓	✗	✓	✓
Kraussl and Tugnetti [21]	✗	✗	✗	✓	✗	✗
Schwiderowski et al. [22]	✓	✗	✗	✗	✓	✗
Hammi et al. [23]	✗	✗	✓	✗	✓	✗
Wang and Nixon [24]	✓	✗	✓	✗	✓	✓
Belk et al. [25]	✗	✗	✗	✓	✗	✓
Heinonen et al. [26]	✗	✗	✓	✗	✓	✗
Fang et al. [27]	✗	✗	✗	✗	✓	✓
This survey	✓	✓	✓	✓	✓	✓

security of token-based systems. Furthermore, while the study [23] discusses the potential of NFTs in the metaverse, it does not explore how these technologies can be integrated with DeFi protocols. Our research paves the way for a future where virtual assets within the metaverse can be seamlessly integrated with DeFi lending, borrowing, and trading platforms. Finally, unlike Belk et al. [25], who focus on consumer motivations for fractional ownership of NFTs, we address the technical challenges associated with implementing such ownership models on the Bitcoin blockchain using Ordinals and Inscriptions. The comparison between our survey and existing surveys is provided in Table 1.

1.3. Our contribution

As discussed previously, the limitations in existing literature highlight the need for a more comprehensive and collective understanding of Ordinals, Inscriptions, and BRC tokens. Therefore, this survey bridges this gap by exploring the collective impact of these protocols on Bitcoin's evolution. We particularly focus on the novel functionalities introduced by BRC-20 and other related protocols, which expand Bitcoin's transactional capabilities and diversify its use cases. Our research contributes to the field in several ways. First, we provide a thorough examination of BRC protocols to explore their technical specifications and functionalities. Second, we offer a detailed survey of existing applications and services that leverage these protocols to showcase their practical applications within the Bitcoin ecosystem. Third, we critically analyze the limitations associated with BRC protocols and discuss potential alternative approaches. Finally, we pinpoint key research challenges that warrant further exploration and propose promising avenues for future research in this domain. *To the best of our knowledge, this study represents the first comprehensive exploration of the synergistic relationship between Ordinals, Inscriptions, and BRC protocols within the Bitcoin blockchain ecosystem.*

The primary contributions of this paper are as follows:

- **Comprehensive analysis of BRC and related token protocols:** We provide a detailed analysis of the development and influence of various BRC token protocols (e.g., BRC-20, BRC-100, and BRC-420) alongside other standards such as ARC-20 and ORC-20. This analysis explores how these protocols have played a crucial role in advancing the Ordinal innovations and their unique contributions to the Bitcoin overall network's performance and compatibility.
- **Integration with Bitcoin's UTXO model:** We explore into the specific functionalities and operational methods of these protocols to emphasize their integration with Bitcoin's UTXO model. This includes exploring essential operations such as deployment, token minting, and transfer to highlight the novel approach of BRC implementations within the Bitcoin network.
- **Applications and their impact:** We describe the diverse applications and services that leverage BRC protocols to showcase their compatibility and adaptability across various sectors. This includes a comprehensive analysis of their impact on areas such as DeFi, wallet technologies, bridging mechanisms, launchpad applications,

and Inscription-based services. Through these examples, we aim to demonstrate the extensive influence and versatility of BRC protocols in different aspects of blockchain technology.

- **Research challenges and future directions:** We identify and discuss significant research challenges and future directions associated with BRC protocols. This includes addressing scalability limitations, protocol complexity, inscription numbering schemes, user experience considerations, infrastructure support, real-world asset (RWA) tokenization, and the long-term stability of these protocols. Additionally, we examine potential alternatives to BRC protocols to offer a perspective on potential solutions and improvements in the field.

1.4. Organization of this study

The remainder of the paper is organized as follows. In Section 2, we present a brief overview of blockchain, DeFi, Web3, and Bitcoin, along with their associated features such as UTXO, SegWit, and Taproot. This foundation is necessary to understand the intricacies of the key concepts in this paper. In Section 3, we explore Ordinals and Inscriptions to illustrate their overview, working principles, and significance in relation to Bitcoin and NFTs. In Section 4, we focus on various types of BRC protocol standards for Bitcoin and provide a detailed analysis of their principles and functionalities. Applications and services utilizing these BRC protocols are comprehensively discussed in Section 5. In Section 6, we address the current challenges and potential solutions related to Ordinals, Inscriptions, and BRC protocols. Finally, we conclude our paper in Section 7, where we summarize the key findings and implications of this study. Note that we use the terms *standards* and *protocols* interchangeably when referring to Bitcoin and BRC standards. This is because both terms accurately reflect the role of BRC-20 and its potential successors in defining token creation, management, and communication methods on the Bitcoin [28].

2. Backgrounds

In this section, we begin with an introduction to blockchain technology, followed by discussions on DeFi and Web3. We then explore Bitcoin's architecture to focus particularly on its UTXO model and its implications for complex applications such as smart contracts and NFTs. Additionally, we provide an overview of Bitcoin's enhancements, such as SegWit and Taproot, which address the limitations of the UTXO model to enhance scalability and privacy. These foundational concepts lay the groundwork for a deeper exploration of Ordinals and Inscriptions in the next section.

2.1. Blockchain

Blockchain technology revolutionizes transaction recording by enabling a decentralized, verifiable, and permanent distributed ledger system [29,30]. Blockchain groups transactions into cryptographically linked "blocks" using hash functions. Leading blockchain platforms, such as Bitcoin [1] and Ethereum [3], rely on consensus protocols like

Proof-of-Work (PoW) and Proof-of-Stake (PoS) to validate and secure the addition of new blocks. These platforms leverage crypto-economic incentives (e.g., auctions [31]) to disincentivize malicious behavior and ensure the integrity of the blockchain. One of the most significant features of blockchain is its immutability. This means that once a block is added to the chain, the transactions it contains cannot be modified or deleted. This guarantees a permanent and tamper-proof record of all transactions. The decentralized structure of blockchain eliminates the necessity for a central authority [32,33]. This mitigates the risk of a single point of failure and improves the system's resilience against different attacks [34]. Blockchain nodes, also known as peers, maintain copies of the ledger and engage in a consensus process to ensure the ledger's accuracy and synchronization across the entire network. The robust architecture of blockchain is essential for enhancing the security and transparency of digital transactions. This is particularly important for applications such as anomaly detection [35], Internet of Things (IoT) [36], and artificial intelligence (AI) [37]. Furthermore, the integration of blockchain technology with AI and machine learning (ML) opens up avenues for innovative solutions [38]. The transparency and security provided by blockchain significantly enhance data integrity, traceability, and trust in decentralized AI systems [39]. This offers new opportunities for decentralized ecosystems, such as DeFi [11], Web3 [40], and Metaverse [41].

2.2. Decentralized finance

DeFi represents a paradigm shift in the financial ecosystem, which leverages blockchain to create open, permissionless, and trustless financial services [11]. Unlike traditional finance, which relies on centralized intermediaries such as banks and financial institutions, DeFi operates through smart contracts deployed on blockchain platforms, primarily Ethereum. The core components of DeFi include smart contracts, decentralized applications (dApps), stablecoins, liquidity pools, and automated market makers (AMM) [42]. DeFi protocols are designed to be interoperable, which allows different dApps to interact and integrate seamlessly. This composability is often referred to as "Money Legos", where users can create complex financial products by combining the DeFi core components. While Ethereum remains the dominant platform for DeFi applications, Bitcoin is also emerging as a viable foundation for DeFi. Protocols such as the Lightning Network (LN) [43] and Rootstock (RSK) [44] facilitate this shift by introducing smart contract functionality to the Bitcoin network. These advancements enable Bitcoin to integrate with the rapidly growing DeFi ecosystem. As a result, Bitcoin's utility extends beyond merely serving as a store of value to include lending, borrowing, and decentralized exchange services.

2.3. Web3

Web3 signifies the evolution of the internet, distinguished by its decentralized framework designed to give users greater control over their data and digital assets [14]. In contrast to Web2, which is dominated by centralized platforms that manage user data, Web3 employs distributed technologies such as blockchain, cryptography, and decentralized networks to give back authority to the users [45]. The central principle of Web3 is data ownership and value expression to empower users with direct control over their digital identities and assets. Key elements of Web3 include dApps, smart contracts, DeFi, and decentralized autonomous organizations (DAOs), all of which operate on blockchain platforms to ensure secure, transparent, and trustless interactions [15]. This ecosystem is further strengthened by distributed technologies, such as the InterPlanetary File System (IPFS) for decentralized storage and P2P networks for communication and data sharing [46].

Blockchain technology serves a fundamental role in the Web3 ecosystem, providing the backbone for dApps and services [40]. Examples of Web3 applications include decentralized exchanges (DEXs) such as Uniswap [47], which enable P2P trading of cryptocurrencies without

the need for centralized exchanges. Additionally, yield farming platforms such as Aave [48] and Compound [49] allow users to earn interest on their crypto assets. Besides, NFT marketplaces like OpenSea [50] facilitate the creation, buying, and selling of unique digital assets. The integration of AI in Web3 further enhances these applications by offering advanced data analytics, optimizing resource allocation, and improving security measures [51]. These examples illustrate how blockchain technology enables decentralized financial services that are more accessible, transparent, and resistant to censorship [40].

2.4. Bitcoin and UTXO

Bitcoin [1], developed by the pseudonymous Satoshi Nakamoto, is recognized as the pioneering blockchain platform and was introduced as the world's first cryptocurrency. Bitcoin is fundamentally based on two key concepts: the Nakamoto Consensus and the UTXO Model. Bitcoin utilizes the Nakamoto Consensus for block agreement, where block generation involves miners solving complex computational puzzles, known as mining. In this mining process, miners determine a nonce value that meets the designated difficulty level. By adjusting this difficulty, the Bitcoin network ensures a consistent block generation pace, approximately one block every ten minutes. When a miner successfully creates a new block, it is disseminated throughout the network. Other miners then append this block to their respective local blockchains. Ideally, the addition of this new block extends the main chain by one block. Nonetheless, network delays sometimes prevent the timely reception of new blocks by all miners.

On the other hand, the UTXO model [52,53], adopted by Bitcoin and other notable blockchain systems [54,55], serves as a fundamental strategy for managing digital currency. In this model, each UTXO signifies a portion of digital currency, tracing a lineage of ownership and transfers through cryptographic techniques. The collective set of UTXOs within a blockchain framework represents all UTXOs, each available for use in future transactions. A user may hold multiple UTXOs, with their total cryptocurrency holdings being the sum of all their UTXOs. This approach effectively mitigates double-spending attacks [56], providing a critical security feature. Conversely, Ethereum [3] employs an account-balance model, similar to traditional banking systems. Ethereum's model features an account-based structure with a transaction count, or nonce, in each account. This helps prevent double-spending and complements the model's straightforward design.

While Bitcoin's UTXO model excels at secure and transparent value transfer, it introduces several limitations for data-intensive applications such as smart contracts, NFTs, digital wallets, data provenance, and supply chain management in terms of data size, complexity, and security. For example, UTXOs are primarily designed for financial transactions, not data storage. Their size is limited to a few hundred bytes, making them unsuitable for storing large or complex data structures. Besides, UTXOs lack inherent mechanisms for representing complex data relationships or conditions. This restricts their use in data-intensive applications requiring advanced logic or programmability. Furthermore, storing large data directly in UTXOs raises security concerns due to potential vulnerabilities with transaction malleability and replay attacks [57]. These limitations leave a gap for exploring alternative approaches like Ordinal and Inscription, which aim to unlock the full potential of Bitcoin's underlying technology for data storage and diverse applications.

2.5. Segregated witness

The Bitcoin protocol processes and authenticates a new block roughly every 10 to 15 min, each containing a defined number of transactions [1]. Historically, Bitcoin transactions incorporate signature data as part of their structure, which is essential for transaction verification. This inclusion introduces a scalability and transaction malleability problem. Segregated Witness (SegWit) [58] is a pivotal upgrade to the

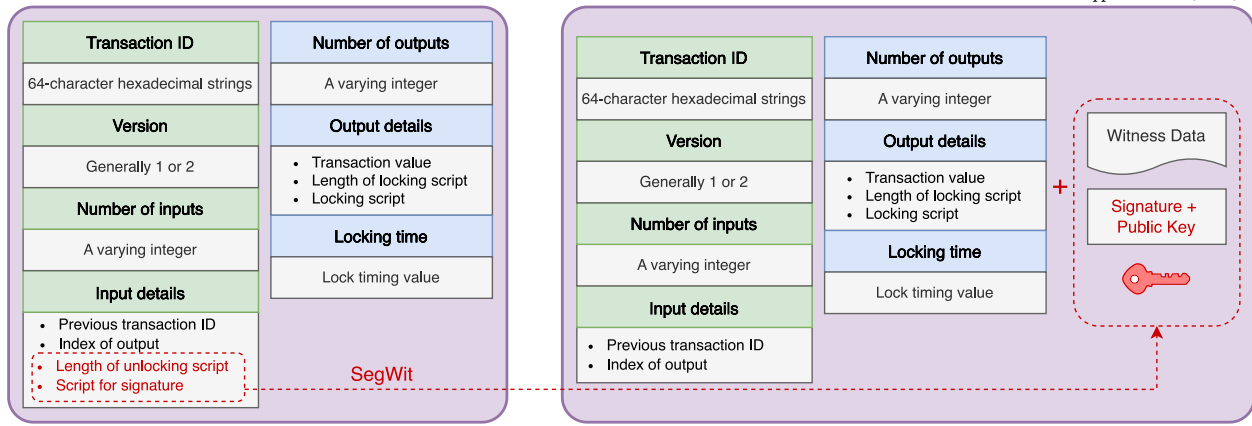


Fig. 1. Enhanced Bitcoin transaction structure with SegWit.

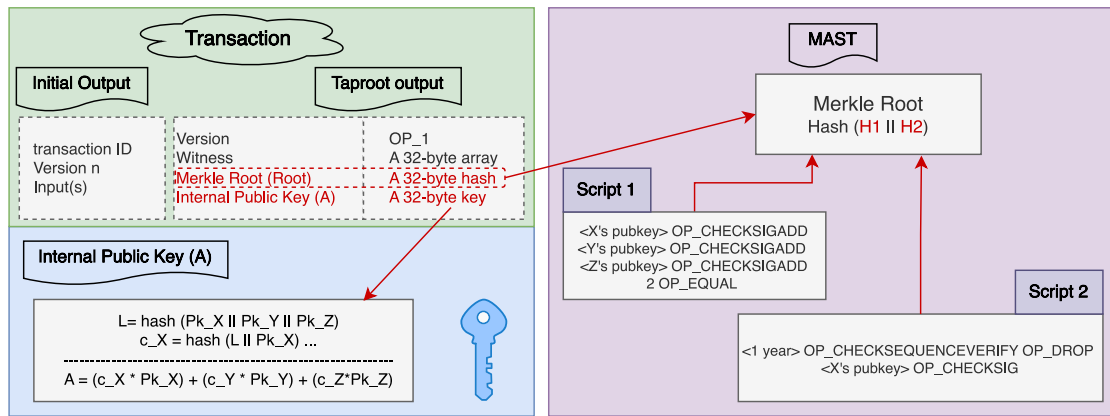


Fig. 2. Taproot transaction with tapscript integration.

Bitcoin protocol, primarily designed to enhance the network's scalability and solve transaction malleability issues [59]. SegWit mitigates these problems by modifying the organization and storage of transaction data within blocks (shown in Fig. 1). It introduces a two-part transaction structure: the main body, excluding the signature, and an additional "witness" component that contains the signatures and scripts. Importantly, the signature data is separated from the transaction data, which influences the transaction ID (txid) calculation. By excluding the witness data from the txid, SegWit ensures the stability of the txid even if the signature data changes. Furthermore, SegWit introduces a "block weight" metric, combining the sizes of blocks with and without witness data. The maximum block weight is established at 4 MB, although the non-witness data remains limited to 1 MB [60]. This adjustment increases the number of transactions each block can carry, thereby enhancing the network's throughput.

SegWit offers several advantages that enhance the functionality of Bitcoin, particularly beneficial for Inscriptions. Firstly, SegWit allows for more transactions to be included in each block by redefining how data is accounted for in the block size limit. This is especially advantageous for data-driven inscriptions, as it provides additional space for these transactions. Secondly, the optimization of block space usage leads to increased transaction efficiency across the network. This efficiency translates into quicker confirmation times and potentially lower transaction fees, which are particularly beneficial for Inscription-based transactions. Lastly, SegWit enhances transaction security by ensuring that changes to the signature data do not affect the txid. This stability in the txid is crucial for maintaining the integrity of references in Inscriptions, as these often rely on unaltered transaction IDs for their operations.

2.6. Taproot

BIP-341 [61], known as Taproot, builds upon BIP-340 [61] by integrating the benefits of Merkelized Abstract Syntax Trees (MAST) [62] and Schnorr signatures [63]. The key innovation of Taproot is to commit a single Schnorr public key in the output that can represent both a single public key transaction and a complex script-based transaction. It introduces a new Taproot output (SegWit version 1) that comprises a signature, a control block, and a script path. Taproot specifies the rules for utilizing the output, which can be done either via a key-path transaction (using a single signature) or a script-path transaction (using one of the scripts committed to in the MAST structure). This approach enhances privacy and efficiency by enabling users to mask complex smart contracts as standard single-signature transactions.

Taproot [64], along with BIP-342 [61], improves scalability by reducing the size of Inscription transactions and enabling batch verification of signatures, which is essential for multi-signature transactions commonly used in Inscriptions. To illustrate the principles of BIP-341 and BIP-342 [61], consider a scenario where a Bitcoin address is controlled by three parties: X, Y, and Z. They create a Taproot output address that allows for flexible transaction conditions using MAST and Tapscript (illustrated in Fig. 2). The transaction conditions are: 1) Any two of the three parties (X, Y, Z) can jointly authorize the transfer of funds, and 2) If the funds remain unspent for a year, party X can unilaterally transfer them (a time-lock condition). Initially, the parties aggregate their individual public keys (Pk X, Pk Y, Pk Z) to form a single Schnorr internal public key A. This aggregate key, combined with the Merkle root derived from two hashed scripts (Script 1 ensures that at least two signatures are provided; Script 2 uses OP_CHECKSEQUENCEVERIFY to

enforce the time-lock and checks for X's signature), forms the Taproot output. When it is time to transfer the funds, the parties can choose a key-path transaction using A for a private transfer, or a script-path transaction, revealing the selected script and its corresponding Merkle proof to meet specific conditions.

3. Two sides of the coin: ordinals and inscriptions

In this section, we explore the innovative concepts of Ordinals and Inscriptions within the Bitcoin network. We start by introducing the Ordinal framework, which assigns unique serial numbers to each satoshi to enhance data storage and support complex applications. Then, we shift our attention to the Inscription mechanism, which allows varied content within individual satoshis (sats) through a Taproot commit-reveal process. We also discuss the significance of these advancements and highlight their key differences to emphasize their unique contributions to the Bitcoin ecosystem.

3.1. Overview of ordinals

The concept of Ordinals originated from Ordinal theory, which was introduced by Casey Rodarmor [65]. Ordinals represent an innovative solution to address limitations associated with Bitcoin's UTXO model [66]. These limitations primarily concern data storage capacity and the ability to support complex applications. It expands the blockchain's capacity for data embedding and advanced functionalities by enabling the creation of Ordinal NFTs directly on the Bitcoin network. Unlike existing NFT solutions on separate platforms, Ordinals leverage the existing transaction system without modifying the core protocol or consensus mechanism. This native integration offers unique benefits, including enhanced security and immutability inherited from the Bitcoin blockchain. While Bitcoin's native unit, the satoshi (sat), is inherently divisible, Ordinal theory allows for individual tracking and identification of each sat within transactions and wallets. Each sat receives a unique serial number based on its creation order within the coinbase transaction and subsequent transaction history.

There are two key aspects of the Ordinal framework. The first is *Serial Numbering*, where each sat is assigned a unique serial number based on its position in the transactions to ensure a distinct identity for each sat. The second aspect is the *First-in-First-out (FIFO) Flow*, a principle where sats maintain their order as they move from input to output during transactions, preserving their sequential integrity. Fig. 3 provides an illustration of the numbering order and flow associated with Ordinals. The left side showcases the minted Ordinals from the Bitcoin genesis block. Ordinal numbering ranges from 0 to 4,999,999,999, corresponding to the initial mining reward of 50 BTC. This visualizes the unique numbering assigned to each sat within the block. The right side depicts the transferred Ordinal process. Inputs such as [sat 1], [sat 2], [sat 3], and [sat 4] are transferred to outputs while maintaining the FIFO order. This mechanism guarantees that the order (or sequence) of sats is maintained throughout transactions. Preserving this sequence facilitates the tracking of sat movement and ownership changes.

Serial numbering and FIFO flow within the Ordinal framework collectively enable the management and tracking of individual sats within the Bitcoin network. To interact with Ordinals, users require specialized wallets equipped to send specific sats to designated transaction inputs. Unlike traditional wallets that treat all sats as fungible (i.e., interchangeable), Ordinal-compatible wallets grant users granular control and tracking capabilities for individual sat units. In addition, Ordinals introduce a novel layer of complexity and utility to the Bitcoin ecosystem. This opens doors for the development of applications that require complex interaction and state management. Functionalities such as complex state management, asset tracking, advanced scripting capabilities, and enhanced metadata storage were previously limited by the UTXO model. Therefore, the Ordinal framework overcomes the UTXO mod-

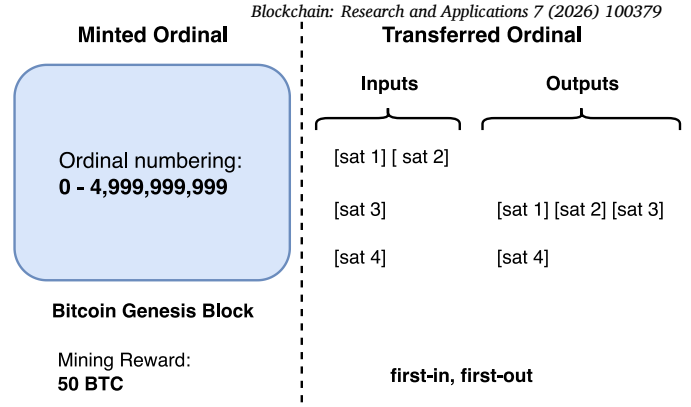


Fig. 3. Ordinal numbering and flow illustration.

el's scripting and scalability issues to unlock new possibilities for digital asset creation and management on the Bitcoin network.

3.2. Significance of ordinals

Ordinals offer a unique set of advantages and trade-offs compared to alternative approaches like sidechains [67,68] and colored coins [69]. Unlike sidechains, which establish separate blockchains with bespoke functionalities, Ordinals leverage the existing Bitcoin network. This allows Ordinals to inherit the security and decentralization of Bitcoin. As a result, they avoid the added complexity and potential security risks associated with maintaining a separate chain. Nevertheless, sidechains [67,68] offer greater flexibility for data storage and scripting. This flexibility facilitates the development of more complex decentralized applications (e.g., Compound [49], Aave [48]). On the other hand, Colored coins [69] represent assets by attaching data to existing UTXOs. Their data capacity and integration with the Bitcoin protocol differ from dedicated token standards. Accordingly, Ordinals operate within the UTXO model but bypass data size limitations by utilizing external storage. This approach provides a more robust and transparent solution for data embedding. It enables novel applications that might be challenging for sidechains and colored coins.

The choice between these approaches depends on the specific needs and priorities of the application. For data-intensive applications within the native Bitcoin ecosystem, Ordinals are a preferable option. This preference is due to their ability to integrate seamlessly with the Bitcoin network, leveraging its inherent security and decentralization. Additionally, Ordinals provide a more efficient method for embedding and managing large volumes of data without the constraints imposed by the native UTXO model. This makes them particularly well-suited for applications that require high transparency and robustness in data handling. Moreover, by utilizing external storage, Ordinals can support a wider range of complex and innovative use cases that might be difficult to implement with either sidechains or colored coins.

3.3. Bitcoin stamps and their distinction from ordinals

The Bitcoin Stamps [70] introduce a novel mechanism for embedding digital assets, such as image-based collectibles, directly onto the Bitcoin. Specifically, the protocol facilitates the creation of permanent digital collectibles and NFTs by leveraging Bitcoin's UTXOs. This design inherently ensures data immutability and resistance to node pruning. Unlike conventional NFT platforms that typically depend on external, off-chain storage solutions, Bitcoin Stamps store asset data entirely on-chain. Consequently, this approach delivers a more robust and decentralized framework for digital art and collectibles. Demonstrating notable adoption, the protocol recorded the creation of over 260,000 Stamps by early 2024, underscoring significant growth relative to other Bitcoin-based token standards [71]. Built upon the established Counterparty protocol [72], Bitcoin Stamps expand Bitcoin's applicability

Table 2
Feature differences between Ordinals and Bitcoin stamps.

Features	Bitcoin ordinals	Bitcoin stamps
Ecosystem	Bitcoin (via SegWit/Taproot)	Bitcoin (originally Counterparty)
Storage location	Witness field (SegWit)	Multi-signature UTXOs (OP_RETURN)
Permanence	Prunable by nodes	Immutable (unprunable)
Transaction cost	Lower (witness discount)	Higher (UTXO bloat, no discount)
Minting flexibility	One inscription per transaction	Supports batch minting
Data size limit	≤4 MB (SegWit limit)	≈7-8 KB (transaction limits)
File formats	Text, images, audio, HTML	SVG, GIF, PNG (24x24px max)
Token standards	BRC-20 (fungible)	SRC-20 (fungible), SRC-721 (NFT)
Security model	Single-signature basis	Multi-signature framework
Primary use cases	NFTs, asset tracking	Long-term storage, secure assets
Adoption status	Established ecosystem	Emerging (OpenStamp, RareStamp)

within DeFi and broader Web3 ecosystems by providing unique mechanisms for asset minting and trading. Technically, the Bitcoin Stamps protocol embeds data into multi-signature UTXOs utilizing the `OP_RETURN` script, which permits limited data storage directly on-chain to ensure that the data cannot be pruned by full nodes.

A pivotal characteristic of Bitcoin Stamps lies in its compatibility with specialized token standards, notably *SRC-20* for fungible tokens and *SRC-721* for NFTs [73]. These token standards enable the direct minting and management of digital assets on the Bitcoin blockchain, emphasizing increased security and longevity. Furthermore, employing multi-signature UTXOs not only ensures enhanced security but also fortifies the data against potential censorship risks. Table 2 summarizes the comparative features of Bitcoin Stamps and Ordinals protocols. The choice between Ordinals and Bitcoin Stamps depends on the application's priorities. Ordinals are better suited for cost-sensitive applications requiring larger data payloads, such as complex NFTs, but face pruning risks. Bitcoin Stamps excel in scenarios demanding permanence, such as digital art preservation, but their higher costs and scalability concerns necessitate careful consideration. Both protocols underscore the evolving role of Bitcoin in supporting diverse data-embedding applications, contributing to its expansion in DeFi and digital collectibles.

3.4. Overview of inscriptions

Inscriptions [65] address some key limitations of Bitcoin's UTXO model, particularly its limited functionality for advanced and diverse applications. This innovative approach allows for the embedding of varied content within individual sats. By doing so, Inscriptions extend Bitcoin's utility beyond its traditional role as a digital currency. Unlike traditional Bitcoin transactions, inscriptions enable the creation and management of unique NFTs directly on the Bitcoin network. This eliminates the need for reliance on auxiliary protocols like sidechains or separate token systems. This innovation overcomes the UTXO model's limitations in handling complex data and managing a wider variety of assets, paving the way for a new generation of blockchain applications. One key advantage of inscriptions lies in their simplified transaction management. Unlike regular Bitcoin transactions, which require precise control over input and output sequences and values, inscriptions leverage the principles of Ordinal theory [65] to streamline the process of sending individual sats.

The structure of Inscription content closely follows the principles of web architecture, a parallel that underscores its innovative approach in the blockchain domain. In web architecture, content is typically organized and accessed through standardized formats, such as MIME types, and is represented in a universally understood medium like byte strings. This similarity in structure allows inscription content to seamlessly interface with traditional web protocols. Consequently, it facilitates the retrieval of Inscription content from web servers, similar to how web resources are accessed. Moreover, creating HTML Inscription enables dynamic integration and modification of content from other Inscriptions. This feature imitates the web's interconnected nature, where resources are embedded and updated in a real-time manner.

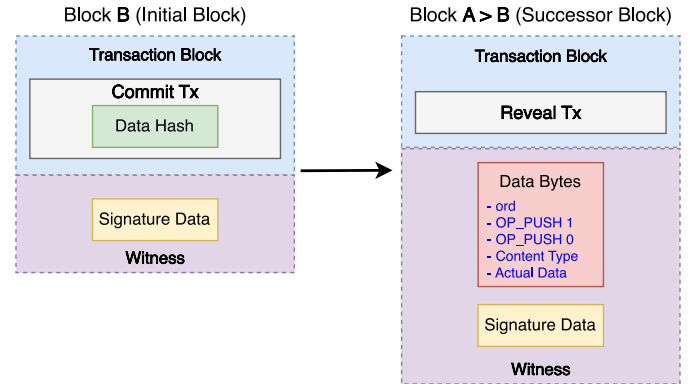


Fig. 4. Two-phase commit-reveal mechanism of inscription.

Inscriptions leverage a two-step commit-reveal process, a mechanism inherent to Taproot scripts. In the first step, a commit transaction generates a Taproot output that references a script containing the Inscription's content. Subsequently, a reveal transaction spends the output from the commit, effectively disclosing the Inscription's content on the blockchain. As depicted in Fig. 4, a transaction within block *B* deposits sats into a script containing the hash of the Inscription data. This hash is stored within the transaction data of block *B*. The witness section holds the signature data necessary for transaction validation. This approach effectively embeds the hash while omitting the SegWit discount due to its placement within the output script. By committing the hash in the initial transaction, the data is secured without immediate revelation, ensuring both privacy and data integrity until the reveal phase. Subsequently, in block *A* (following block *B*), a separate transaction spends the previous output, thereby revealing the actual inscription data. The data bytes themselves comprise several components: the string `ord` signifying the Inscription's specific purpose, followed by `OP_PUSH 1` to indicate that the next data push will contain the content type, and `OP_PUSH 0` to denote that subsequent data pushes will hold the actual content. Large Inscriptions necessitate multiple data pushes, as a single push is limited to 520 bytes. These data bytes reside within the witness section of block *A*, subsequent to block *B*. This placement ensures the Inscription content becomes publicly available while adhering to the SegWit structure, which segregates transaction data from witness data. Finally, this data unlocks the SegWit discount to reside within the input script of a segregated witness transaction.

3.5. Significance of inscriptions

Inscription has become popular in the crypto world for several reasons. First, it allows users to create and trade NFTs on Bitcoin, which are unique digital assets associated with art, music, images, videos, and games to identify the authenticity of the owner [74]. NFTs have gained a lot of attention and value in the crypto space, as they enable new forms of creativity, ownership, and monetization. Inscription enables

NFTs on Bitcoin by attaching metadata to sats that identify and verify the authenticity and provenance of the digital asset [75]. Second, Inscription enhances the programmability and functionality of Bitcoin, as it stores smart contract codes, scripts, oracles, and other logics which are executed on the Bitcoin layer 2 (L2) solutions such as Stacks [76] and LN [43]. Inscription also enables cross-chain interoperability, as it stores data from other blockchains or protocols that are verified and used on Bitcoin. Inscription thus expands the use cases and innovation potential of Bitcoin, as well as its network effects and security [77,78]. Third, Inscription increases the demand and value of Bitcoin, as it creates new incentives and opportunities for users, developers, and miners. Fourth, the emergence of Inscriptions presents a win-win scenario for various stakeholders within the Bitcoin ecosystem. Users gain access to a diverse and rich landscape of applications and services built upon the Bitcoin network. Developers benefit from the inscription mechanism's inherent simplicity and efficiency, allowing them to leverage existing Bitcoin infrastructure and tools for project creation and deployment. Finally, miners experience an increase in transaction fees and rewards as they process and validate the additional data and transactions inscribed on the blockchain [79].

3.6. Inscriptions vs. NFTs: key differences

Inscriptions and NFTs represent distinct methodologies for managing digital assets. For example, inscriptions leverage the Ordinal protocol, which embeds data directly within individual sats. This ensures complete on-chain storage to guarantee immutability and permanence of the data. In contrast, NFTs are compatible with the Ethereum Virtual Machine (EVM) and utilize standards such as ERC-721 [80] and ERC-1155 [81]. These standards often rely on off-chain storage solutions like IPFS [82] and Arweave [83] for the actual digital asset content, with only a token reference remaining on the blockchain itself. Additionally, the on-chain nature of Inscriptions inherently links their limited availability to Bitcoin's finite supply of 21 million coins. In contrast, NFTs can be produced in much larger quantities due to flexible minting policies, potentially leading to market saturation for certain NFT types. Current Inscription minting typically requires a dedicated node or third-party services, with trading occurring through NFT marketplaces. Conversely, NFTs on programmable blockchains offer more seamless integration and can often be minted directly through web interfaces. Furthermore, inscriptions lack inherent support for on-chain royalty models, indicating a lack of automatic mechanisms to provide a percentage of sales back to the original creator. In contrast, many NFTs on platforms like Ethereum use smart contracts to include on-chain royalty models, ensuring ongoing compensation for creators. Integration of Inscriptions with other platforms is challenging due to Bitcoin's scripting limitations, and their energy consumption is high due to Bitcoin's PoW consensus. NFTs are confined to specific ecosystems with varying energy footprints based on the platform's consensus algorithm.

4. Protocol standards for bitcoin: Frameworks and methodologies

In this section, we present a comprehensive technical overview of BRC token standards in terms of their definition and working principle. BRC tokens have been instrumental in advancing Ordinals and Inscriptions. We explore the evolution of Bitcoin protocols to focus on the pivotal roles of BRC-20, BRC-100, BRC-420, and other relevant standards from the perspective of Bitcoin's transactional landscape. We investigate each standard's unique contribution to the functionality and interoperability within the Bitcoin network. Fig. 5 illustrates the timeline of Bitcoin protocol developments, beginning in February 2022 with the inception of Ordinal numbers. This is followed by the introduction of the BRC-20 standard in March 2023, which advanced Bitcoin's tokenization framework. Subsequent enhancements include the ARC-20 standard in July 2023 and the BRC-100 and BRC-420 standards, both introduced in

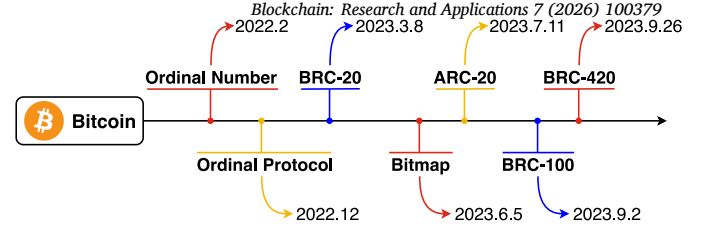


Fig. 5. Evolution timeline for Bitcoin token standard.

September 2023. These developments collectively refine and extend Bitcoin's functionality and application potential. A comparison of different Bitcoin standardized token protocols is presented in Table 3.

4.1. BRC-20 protocol

Building upon the foundation of Ordinal and Inscription, BRC-20 (Bitcoin Request for Comment 20) defines an experimental standard for fungible tokens on the Bitcoin ecosystem. BRC-20 facilitates the creation, transfer, and management of fungible tokens directly on individual sat inspired by Ethereum's ERC-20 [84]. This innovative approach utilizes JSON data embedded within the sats to encode token metadata and functionality, enabling features like decimalization, supply control, and customizability. The BRC-20 standard builds upon the foundation of ERC-20, a widely adopted token standard on the Ethereum blockchain. In addition, non-fungible assets within the broader Inscription framework are represented through Ordinal Inscriptions and NFT-oriented standards such as BRC-69 [90]. This creates a distinctive hybrid standard, tailored to Bitcoin's environment. Unlike Ethereum's model, BRC-20 operates without smart contracts and is specifically designed to navigate the constraints of Bitcoin's UTXO data structure. The development of BRC-20 marks a significant stride in enhancing Bitcoin's capabilities and broadening its use cases in the digital currency landscape.

Some of the well-known BRC-20 tokens are ORDI [85], PEPE [85], MEME [85], PIZA [85], and DOMO [85]. For example, ORDI [85] stands as the first governance token within the Ordinal protocol framework. It gives its holders the authority to influence the direction of the Ordinal ecosystem through voting on various proposals and settings. Similarly, PEPE [85] is widely recognized as an internet meme depicting a frog with an array of expressions. This token provides a platform for the creation and trade of frog-themed NFTs on Bitcoin. MEME [85], on the other hand, facilitates the creation and trade of meme-centric NFTs on the Bitcoin network, while PIZA [85] represents a novel means to transact pizza purchases using Bitcoin. Additionally, the DOMO [85] token bears the name of the BRC-20 architect and serves a unique role within the ecosystem.

4.1.1. BRC-20 implementation on bitcoin

The BRC-20 implementation on Bitcoin is a novel approach that integrates the stateless UTXO model of Bitcoin with the stateful token concept of BRC-20. This is achieved through the use of Ordinal to embed specific information within sats, enabling the tracking and transfer of tokenized assets. The implementation primarily involves three key processes: deployment, minting, and transferring of tokens. Each operation begins with an on-chain action, which triggers a corresponding off-chain update to maintain a parallel state reflective of the blockchain events. This dual-layer structure ensures both the immutability and transparency of the blockchain while enabling the flexible management of token states off-chain.

Deployment process The deployment process serves as the foundation for creating a new BRC-20 token. It begins with an on-chain action where a sat is inscribed with essential information such as the protocol name, operation type, token name, and token limits. This creates a unique identifier for the token on the Bitcoin network. Subsequently, an

Table 3
Comparison of BRC token protocols.

Criteria	BRC-20	BRC-69	BRC-100	BRC-420	ORC-20	ARC-20
Ease of use	Moderate, JSON-based	Simple, Ordinal-focused	High, complex governance and state models	Moderate, modular development	Moderate, BRC-20 compatible	Moderate, Atomicals protocol
Security	High, UTXO-based	High, immutable and transparent	High, UTXO and state models	High, secure usage rights and royalties	High, double-spending prevention	High, transparent transactions
Scalability	Moderate, depends on network fees	High, optimized for block space	High, large-scale support and management	High, diversified	High, adaptable and scalable	High, platform integration
Complexity	Moderate, off-chain state	Moderate, deploy and compile operations	High, inheritance and nesting	High, modular with royalty function	Moderate, multiple operations	Moderate, various digital assets
Transaction model	Off-chain state	Block space usage	State machine models	Integrate with PSBT	various operations	UTXO-based, traceable transactions
Interoperability	Limited to Bitcoin	High, support recursive Inscriptions	High, compatible with extensions	High, broad adoption	High, BRC-20 backward compatible	High, interoperable with Ordinals
Token standards	Fungible	Non-Fungible	NFTs with computing	Complex assets and royalties	Fungible	Interchangeable
Use Cases	Token management	Collect Ordinals	App with computing support	Metaverses, gaming asset management	Token creation	Digital identities, NFTs, domains

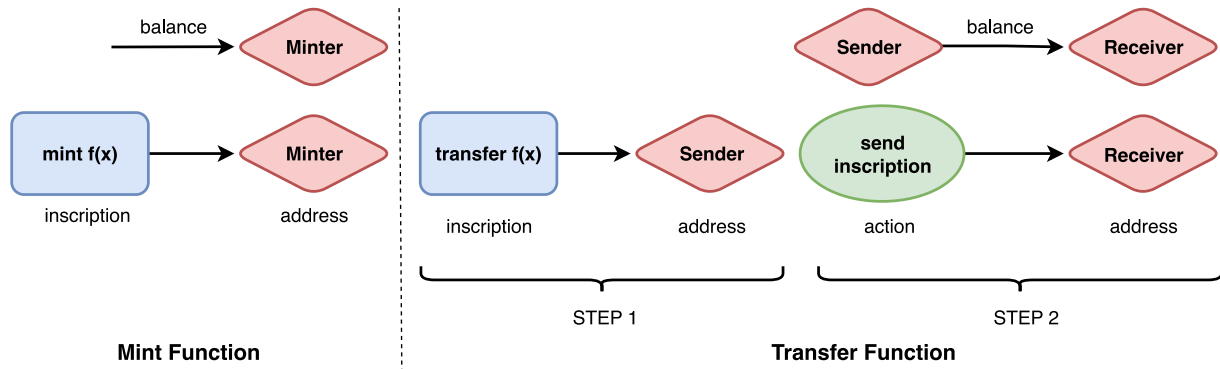


Fig. 6. Minting and transfer process of BRC-20.

off-chain process checks for the existence of a state for the token. If no state exists, a new state is initialized with predefined token properties and account balances. This dual process ensures the token's integration into the Bitcoin network while setting up the necessary infrastructure for its management and distribution. The mint and transfer operation (shown in Fig. 6) of BRC-20 is presented as follows:

Minting process Minting new tokens is a crucial functionality of the BRC-20 implementation. The process initiates with an on-chain Inscription of a sat to signify the mint operation. This type of inscription includes details such as the protocol name, operation type, token name, and the amount being minted. Afterward, an off-chain process takes responsibility for verification. This verification involves a three-step approach. Step-1 ensures that the token's state adheres to established validity criteria. Step-2 guarantees that the individual minting does not exceed predefined limits, and Step-3 confirms that the total minted tokens remain within the maximum allowed. Upon successful validation, the minter's balance in the off-chain state is updated to reflect the new token quantity, seamlessly integrating the on-chain event with the off-chain token management system.

Transfer process The transfer operation is designed to facilitate the exchange of tokens between accounts. This operation is slightly more complex as it involves two distinct on-chain transactions. Initially, a sat is inscribed with information including the protocol name, operation type, token name, and the amount being transferred. This inscribed sat

is then transferred to the sender, who must immediately create a subsequent transaction to forward it to the recipient. In parallel, an off-chain update verifies the sender's balance and, upon validation, adjusts the balances of both sender and recipient accordingly. This ensures a secure and verifiable transfer of tokens, bridging the on-chain actions with off-chain account balances.

4.1.2. BRC-20 operation on bitcoin

The BRC-20 operation on Bitcoin leverages the Partially Signed Bitcoin Transaction (PSBT) [86] standard to facilitate the trading of tokens. This method enhances the flexibility of unsigned transactions and allows multiple signatures. In this framework, a seller first inscribes a sat to embed important transaction details such as the protocol, operation type, token name, and amount. The Inscription is then integrated into a PSBT, setting the groundwork for a trade. The PSBT is published in a marketplace to invite potential buyers. A buyer then finalizes the transaction by contributing the necessary Bitcoin amount and receiving the inscribed sat. Following this, the transaction is broadcast to the Bitcoin network to ensure its inclusion in a future block and to finalize the trade. Concurrently, an off-chain process updates the token balances of both parties, reflecting the new state of affairs post-transaction. This dual-layer approach intertwines the decentralized, immutable nature of blockchain with efficient off-chain management to ensure a seamless and secure trading experience.

Table 4
Feature differences between BRC-20 and ERC-20.

Features	BRC-20	ERC-20
Ecosystem	Bitcoin	Ethereum
Token management	Ordinal and Inscription for tokenization	Automated via smart contracts
Fungibility	Interchangeable; one-to-one trading	Interchangeable; application integration
Transaction cost	Higher due to Inscription design	Variable, higher during congestion
Scalability	Affected by transaction fees	Layer-2 scaling for throughput
Primary use cases	Remittances, DeFi	DeFi, NFTs
Innovation	Under development; focus on DeFi features	Widely adopted; focus on scalability

Signature for transaction The PSBT standard, defined in BIP-174 [86], plays a crucial role in facilitating the signing process for BRC-20 transactions on the Bitcoin network. This standard streamlines transaction signing, especially in scenarios involving multiple participants. A PSBT is constructed by specifying the UTXOs to be spent and the desired outputs of the transaction. It then includes the minimal information required for each UTXO to enable independent signature generation by various programs, such as those used by multi-signature wallets. Subsequently, these partially signed PSBT copies are combined to form a single, fully signed transaction ready for broadcast on the Bitcoin network. The PSBT workflow underscores the collaborative and secure nature of transaction signing within the Bitcoin ecosystem. It ensures the integrity of the transaction and cooperative trust among participating parties by requiring their mutual agreement.

Trading cycle for transaction The trading cycle for BRC-20 transactions on Bitcoin is a structured process beginning with the seller's operation. Here, the seller inscribes a sat to signal a transfer of BRC-20 tokens, embedding details like the protocol, operation type, token name, and amount. This inscribed sat is then incorporated into a PSBT, with the seller setting a starting bid. The PSBT is subsequently published to a marketplace, inviting buyers to review and potentially finalize the transaction. When a buyer agrees to the terms, they complete the PSBT by supplying the required Bitcoin, in exchange for the inscribed sat. Finalizing the PSBT involves broadcasting it to the Bitcoin network, ensuring its inclusion in the blockchain and the immutability of the transaction. Simultaneously, an off-chain update reflects the updated token and Bitcoin balances of both the buyer and seller. This cycle not only facilitates the trading of tokens but also seamlessly integrates on-chain actions with off-chain state management, ensuring a secure and efficient transaction process.

4.1.3. Differences with ERC-20

Table 4 represents the differences between BRC-20 and ERC-20 in terms of features such as blockchain ecosystem, token management, fungibility, transaction costs, scalability, primary use cases, and innovation outlook. BRC-20 tokens operate on the established Bitcoin network and utilize Inscription for creating and managing tokens. This leads to higher transaction costs and impacts scalability. In contrast, ERC-20 tokens are based on the Ethereum blockchain and employ automated smart contracts for management, facing variable fees that rise during network congestion but are mitigated by Layer-2 scaling. Both token standards offer fungible assets, allowing for straightforward one-to-one trading and application integration. The BRC-20 tokens are mainly used for P2P transactions and remittances, and are exploring their potential within DeFi platforms, with ongoing development aimed at enhancing DeFi-related features. Meanwhile, ERC-20 tokens are integral to exchanges, DeFi applications, and NFT marketplaces, with a focus on addressing scalability to maintain their wide adoption.

4.1.4. Token burning in BRC-20

One of the key limitations of the BRC-20 token standard is the absence of a built-in burn function, a feature commonly found in the ERC-20 token standard on Ethereum [87]. In Ethereum, the `burn()`

function allows tokens to be destroyed, effectively reducing the circulating supply. This functionality is often used in deflationary models, where tokens are periodically burned to decrease total supply and potentially increase the value of remaining tokens. In contrast, BRC-20 tokens do not have a native method for burning tokens. This limitation arises from Bitcoin's restricted scripting capabilities, which do not support the execution of complex functions like token burning directly on-chain. As a result, the BRC-20 protocol operates without an automated burning mechanism, making it quite different from Ethereum-based standards. The Bitcoin community has suggested several workarounds to address this issue [88]. One proposed method involves sending tokens to a specific address with no private key, essentially *locking* the tokens in an unspendable state. This approach simulates the burning of tokens but does not provide a formal burn function within the protocol itself. While this method is widely acknowledged by the community, it lacks standardization and is not a widely accepted feature in BRC-20 token implementations. In the future, there are ongoing discussions and proposals aimed at enhancing the BRC-20 protocol. For example, the introduction of the Runes protocol [89], a more advanced token standard for Bitcoin, includes a built-in burn function. In the Runes protocol, tokens can be burned by setting a specific `Runestone.burn` attribute to `true` and spending the corresponding transaction output. This addition provides a standardized and formal mechanism for token burning, which is a significant step forward in addressing the current limitations of the BRC-20 standard.

4.2. BRC-69 protocol

BRC-69 [90] is an innovative standard for NFT collections on Bitcoin, utilizing recursive Inscription [91] to optimize block space usage. This standard is designed to manage the rising costs associated with the growing popularity of inscribing data into these collections. It achieves over 90% efficiency in block space utilization through a unique process involving the on-chain Inscription of images, collection deployment JSON, collection compiler JavaScript, and the asset with the mint operation. BRC-69 also introduces features like pre-reveal collection launching and on-chain reveals, with images automatically rendered on Ordinal explorers, reducing the need for external actions. This significantly enhances the efficiency of Bitcoin block space, making it more appealing for creators to launch their ideas on the blockchain.

BRC-69 operates on optimizing the Inscription process for NFT collections. It involves inscribing essential components directly onto the Bitcoin, which works in parallel to create and manage digital assets efficiently and with less block space consumption.

Compilation process The compilation process in BRC-69 leverages a JavaScript Inscription to define the logic for rendering final assets. This Inscription, known as a *Compile Inscription*, employs a recursive approach. It references the *Deploy Inscription* to retrieve the attributes associated with specific Inscription IDs. This mechanism enables customization of rendering features for individual collections, capitalizing on JavaScript's flexibility to facilitate the creation of diverse and dynamic assets.

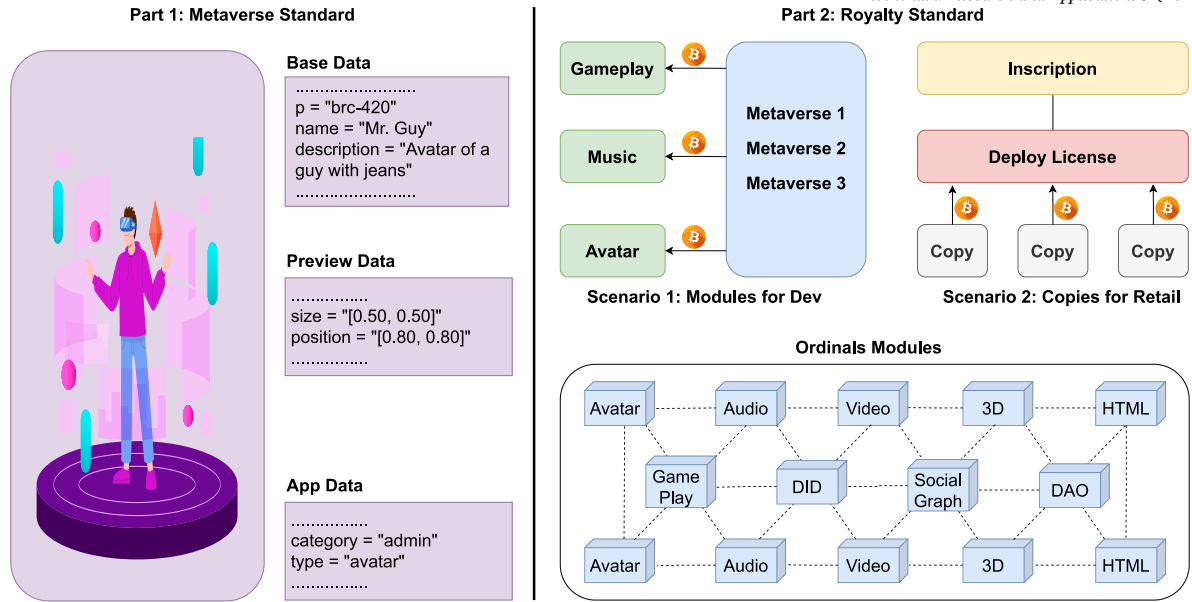


Fig. 7. Protocol standards for BRC-420 Protocol.

Minting process The minting process in BRC-69 is executed through an HTML-type Inscription, which concisely stores the index of traits used to generate the final asset. This points back to the Compile Inscription in a single line, enabling any front-end with Recursive Inscription capabilities to automatically render the image using on-chain data. The streamlined process ensures efficient use of Bitcoin block space while maintaining the integrity and uniqueness of each minted asset.

4.3. BRC-100 protocol

Built upon the principles of Ordinal theory, BRC-100 [92] is an advanced decentralized computing protocol within the Bitcoin ecosystem. A key differentiating factor is its protocol inheritance feature. This allows developers to create extension protocols that inherit properties, operations, and computing capabilities from a parent protocol. BRC-100 further expands functionality through nesting, enabling the creation of child applications under a parent application, each with its own unique ticker symbol. The protocol is separated from the traditional UTXO model as it introduces a state machine model. This allows applications and addresses to have states, managed through specific operators.

Five key parameters define the operational framework of BRC-100 applications: (1) *extend* determines protocol inheritance, (2) *upgradeFrom* identifies upgradable protocols, (3) *openAsChild* and (4) *onlyChild* define deployment permissions for child applications, and (5) *stopable* indicates whether an application can be stopped. These parameters provide developers with precise control over the functionality and lifecycle management of their applications. For example, the *extend* parameter enables seamless integration of new features by inheriting from established protocols to promote innovation while maintaining stability. The *upgradeFrom* parameter ensures a smooth transition to newer versions to keep applications current with the latest technological advancements. Permissions specified by *openAsChild* and *onlyChild* offer flexibility in application deployment to support hierarchical organization and management. Lastly, the *stopable* parameter ensures applications can be safely terminated when necessary to enhance security and resource management. Furthermore, BRC-100 leverages a complementary governance protocol, BRC-101 [93], to facilitate on-chain management of applications. The working principle of BRC-100 is illustrated as follows:

The BRC-100 protocol operates on the foundation of creating, managing, and governing applications with computing capabilities on the Bitcoin network. These applications act as tokens defined by the pro-

tol, and their operations are governed by the protocol's rules and extensions. The working model of BRC-100 is divided into three parts:

Deployment process The deploy operator is essential for deploying applications on the BRC-100 protocol. Based on the *openAsChild* and *onlyChild* parameters, applications are launched directly using the governance protocol BRC-101. These parameters dictate the permissions for deploying child applications. The *stopable* parameter indicates whether an application is terminated on the post-deployment stage.

Minting process The minting process for BRC-100 applications involves creating tokens within the constraints set by the application's protocol. The number of tokens that are publicly minted is predefined, ensuring adherence to the application's maximum settings. The mint operations are designed to increase the token's supply within the rules and parameters of the application, reflecting the application's state and balance.

Burn process The burn process of BRC-100 involves the reduction of a token's circulating supply. Tokens are burned publicly or according to a whitelist to align with the application's rules. These operations facilitate the management of tokens' lifecycle and are integral to applications' state changes to ensure consistency across the BRC-100 ecosystem.

4.4. BRC-420 protocol

BRC-420 [94] redefines traditional digital asset management within the Metaverse [95] and Web3 [13] ecosystems, targeting on-chain gaming solutions. These solutions represent a novel paradigm in game development, where all game aspects, from rules and assets to player activity, are immutably stored on a blockchain [96]. As shown in Fig. 7, the BRC-420 protocol comprises two core standards: (1) the Metaverse Standard, which structures assets through base, preview, and application-specific data to support reusable and composable resources; and (2) the Royalty Standard, which enables licensed reuse and distribution of inscriptions while enforcing royalty payments to the corresponding rights holder. Key features of BRC-420 include:

- **Open-source creation of complex digital assets:** This empowers creators to design intricate in-game assets and virtual experiences.
- **Integration of usage rights with royalties:** BRC-420 facilitates a sustainable creator economy by enabling creators to earn royalties on the usage of their assets.

- **Enhanced creator economy within digital environments:** BRC-420 facilitates a more robust and rewarding environment for content creators in the Metaverse and Web3.

BRC-420 offers a wide range of applications to enable functionalities, such as licensing in-game elements within the Metaverse and streamlining digital asset management across diverse virtual platforms.

The working principle of BRC-420 revolves around asset creation and management, coupled with a unique royalty system linking usage rights to creators' income. The roles in this system include the Owner, who has legal ownership of the Inscription; the Deployer (or Receiver), responsible for deploying BRC-420 and receiving royalties; and the Minter, who creates new content based on the original Inscription.

Deployment process In BRC-420, deployment involves inscribing a deploy function to an Ordinal compatible wallet, specifying the royalty fee during this phase. The protocol allows for the deployment of BRC-420 on any owned Inscription, with the deployer becoming the receiver of the royalty fees. Most importantly, each Inscription undergoes only one BRC-420 deployment, and any attempts at subsequent deployments are invalid, regardless of the Inscription's transfer between accounts.

Minting process The minting process under BRC-420 permits the minter to inscribe recursive content following the parameters set in the deploy function, including the payment of a royalty fee to the deployer. The mime type of the recursive Inscription must match that of the original, and the content adheres to a recursive style referencing the original Inscription ID. During this process, the correct amount of Bitcoin must be sent to the deployer as a royalty fee.

4.5. ORC-20 protocol

The ORC-20 token standard [97] represents an innovative approach to enhancing the functionalities of the BRC-20 standard for Bitcoin Ordinals. Built with backward compatibility in mind, ORC-20 prioritizes scalability, security, and the elimination of double-spending vulnerabilities. This standard aims to facilitate wider adoption by offering functionalities beyond BRC-20's capabilities. These include the deployment of new tokens, minting, sending, and the ability to cancel or modify portions of transactions. Additionally, ORC-20 addresses limitations inherent to BRC-20, such as the immutability of token supply after deployment and restricted naming conventions. It further strengthens transaction security and offers greater flexibility. However, it is crucial to acknowledge that ORC-20 remains an experimental project. As such, the value and utility of tokens created under this standard are not yet guaranteed.

The ORC-20 standard operates on a fundamental principle that prioritizes efficient, secure, and adaptable token transactions within the Bitcoin ecosystem. It leverages a combination of event-based operations and a flexible key-value pair inscription system to enhance user experience and reduce blockchain space consumption.

Event-based process ORC-20 defines a set of event-based operations that govern the lifecycle of tokens created using this standard. These operations include deploy, mint, send, cancel, and upgrade. Each event plays a critical role, encompassing token creation, distribution, transfer, and modification. This event-based approach offers granular control over token behavior while maintaining a structured and flexible framework for managing token functionality.

Inscription process The ORC-20 token standard introduces a flexible key-value pair inscription system, primarily utilizing JSON format. This system allows for significant customization and adaptability. Users can define various constraints, behaviors, or operations associated with the tokens. This approach offers a dynamic and responsive framework compared to previous standards, enabling a wider range of applications and use cases.

Ownership and upgrade process Ownership of ORC-20 tokens is established through the Inscription of a deployment event on the Bitcoin blockchain. This event grants the owner the right to implement future upgrades to the token's functionality. This built-in upgrade mechanism allows ORC-20 to adapt to evolving needs and technological advancements, facilitating its long-term applicability within the dynamic landscape of digital tokens on the Bitcoin network.

4.6. ORC-CASH protocol

ORC-CASH [98] is a streamlined token standard designed for UTXO-based blockchains, specifically built on the Ordinals protocol [97]. It adheres to both the BRC-20 and ORC-20 standards, leveraging the existing functionalities while potentially offering additional benefits. The core design philosophy of ORC-CASH aligns with traditional cash systems, prioritizing security by leveraging the full power of the network's hash function. This ensures immutability and transparency of all on-chain transactions. Additionally, ORC-CASH facilitates versatile deployments, striking a balance between stability and flexibility to accommodate future growth. The working principle of ORC-CASH is presented as follows:

ORC-CASH operates on a principle that emphasizes secure and free transferability of tokens that act as cash bills within the network. The protocol ensures that every transaction record is kept on-chain, allowing for transparent and immutable indexing. It supports multi-purpose deployments with a focus on flexibility and expandability, fostering an environment that is conducive to project development and community collaboration.

Deployment process The deployment of a new ORC-CASH token involves defining a unique ticker with specific attributes such as maximum supply, mint limit, and decimal precision. A deployer Inscription represents ownership of the token, with the capacity for future upgrades if not disabled. Deployments are made multiple times under the same ticker name but must have different Token IDs, with the deployer holding rights to modify token specifications.

Minting process Minting in ORC-CASH allows the creation of token balances associated with a deployed token. It adheres to set limits, ensuring that minting does not exceed predefined thresholds. Once minted, tokens are represented by Inscription that function as cash bills, maintaining their value across transactions until they are burned or split into smaller denominations.

Send process Sending ORC-CASH tokens to another wallet requires executing a send operation that specifies the amount to be transferred. This operation relies on the Token Credit system, where credits are obtained by sending valid cash bills to a Burn Wallet, acting like a universal ATM, enabling the creation of additional valid cash bills with varying denominations.

Upgrade process The upgrade process for ORC-CASH tokens involves modifying predetermined specifications associated with the token, such as its ticker symbol, maximum supply, and mint limit. This upgradability feature is available only if it hasn't been permanently disabled by the Deployer, the entity responsible for deploying the token on the blockchain. Importantly, any upgrades implemented must adhere to existing limitations to safeguard the integrity of the ORC-CASH token's economic model.

Sell process Selling ORC-CASH Token Credits involves setting up a sell operation with parameters like amount, price, and expiration time. This operation turns Token Credits into offerings that are purchased by other users, with immutable records ensuring the traceability and security of the transaction. Sell orders are not canceled once activated and will auto-expire if not fulfilled within the predefined timeline.

Table 5
Lists of applications and services for the BRC ecosystem.

Types	Application/Service
DeFi	Dova [100], Trac [101], ALEX [102], Liquidium [103], BitX [104]
Wallets	UniSat [105], Xverse [106], Leather [107], Ordinal wallet [108]
Bridges	MultiBit [109], OrdBridge [110], OrdiZK [111]
Launchpad	TurtSat [113], OrdMint [114], KoiPond [115]
Inscription	OrdinalsBot [116], Magic Eden [8], Gamma [117], OrdinalHub [118], Ordinal Punks [119]
Stablecoins	bitSmiley [120], BitStable [121]

4.7. ARC-20 protocol

ARC-20 [99] is another protocol within the BRC standards that emerges from the Atomicals protocol [99]. ARC-20 leverages colored coins, a technique that encodes metadata onto the Bitcoin blockchain to represent real-world or digital assets [69]. This approach differentiates ARC-20 from other token standards by enabling the creation of unique tokens directly on Bitcoin. A key feature of ARC-20 is its built-in naming system, which guarantees token uniqueness and simplifies the creation and management of diverse digital assets, including NFTs, digital identities, and domain names. Additionally, ARC-20 offers functionalities such as Graphics Processing Unit (GPU) mineable tokens, enabling efficient token distribution, and support for asset hierarchies through recursion. Furthermore, it emphasizes user-friendliness with interfaces catering to both technical and non-technical users. Beyond these features, ARC-20 promotes streamlined transactions, potentially improving scalability within the Bitcoin ecosystem. Notably, it aims to reduce reliance on third-party intermediaries, aligning with the principle of decentralization that Bitcoin advocates. The working principle of ARC-20 is given below:

ARC-20 operates on the principle of utilizing Bitcoin’s UTXO feature for transparent and traceable transactions, enhancing trust and reliability in the management of digital assets. This protocol allows for the creation, transfer, and enhancement of various digital assets, reinforcing Bitcoin’s role as a hub for digital authenticity.

Transfer process The transfer operation in ARC-20 involves allocating input values to outputs, adhering to a FIFO rule, and bypassing unspendable outputs. In cases where outputs are not fully allocated, a best effort assignment is attempted. Additionally, ARC-20 incorporates a swap transfer operation where a seller signs an input containing an ARC-20 Atomical and an output to receive sats. This allows the buyer to add inputs for funding and receive the ARC-20 tokens, ensuring flexibility and security in transactions.

Minting process The minting process within the ARC-20 framework manage the creation of “Subrealms”. These Subrealms are digital constructs established within a primary realm on the Bitcoin network through the Atomicals protocol [99]. They play a crucial role in facilitating and tokenizing community-based structures. Subrealms offer multifunctional capabilities, including social media organization, identity verification, and loyalty reward system administration. In terms of ARC-20, the minting rules are embedded within the Subrealm data structure. This allows for the creation of Subrealms with varying lifespans and under specific conditions. These conditions involve computational effort or payment in sats. A dedicated file outlining the minting requirements ensures a structured and controlled approach to creating new digital domains within the ARC-20 framework.

5. Applications and services for BRC protocols

In this section, we explore the diverse applications and services built on top of BRC protocols. These protocols play a critical role in expanding the functionality and performance of the Bitcoin network. The emergence of these applications and services highlights the dynamic and evolving nature of the BRC ecosystem. This further solidifies its growing

influence within the broader blockchain landscape. Table 5 provides a list of applications and services that utilize BRC protocols, with a primary focus on BRC-20.

5.1. Application ecosystem for BRC protocols

Here, we discuss all the relevant applications, including DeFi, wallet services, bridging applications, and launchpads, each contributing to the robust expansion and integration of Bitcoin’s capabilities.

5.1.1. DeFi applications

The Ordinal and Inscription-based DeFi ecosystem utilizes the BRC token standard to highlight its dynamic innovations. For example, Dova protocol [100] integrates the MultiBit bridge [109] for seamless BRC-20 token transfers and collateralized lending on Ethereum. On the other hand, Trac Systems [101] act as a Bitcoin indexer and oracle to enable secure data access for diverse operations. Trac [101] introduces a multi-asset meta-protocol for Ordinal to support complex functions like staking and liquidity pools directly on Bitcoin. Moreover, Alex [102] bridges Bitcoin’s L1 and L2 to enhance DeFi capabilities through the smart contracts from Stacks [76]. Liquidium [103] offers a novel Ordinal-based lending approach on Bitcoin, with a peer-to-peer protocol using Ordinal as collateral. BitX [104] addresses Bitcoin Ordinal’s challenges with a backend router, improving liquidity in the BRC ecosystem. These platforms collectively signal a significant leap in Bitcoin-based DeFi, pushing boundaries beyond conventional uses.

5.1.2. Wallet applications

Wallet applications, including UniSat [105], Xverse [106], Leather [107], and Ordinal Wallet [108], offer support for BRC-20 tokens on the Bitcoin network. These wallets expand the functionality of Bitcoin by enabling users to interact with this new token standard. Xverse, for example, is a cross-platform Web3 Bitcoin wallet specifically designed for BRC-20 tokens [106]. It provides users with a seamless experience across Android, iOS, and Chrome extensions. Beyond storage and transfer of BRC-20 tokens and Bitcoin NFTs, Xverse facilitates integration with Alex [102] for DeFi trading. On the other hand, Leather [107] highlights the significance of BRC standards by emphasizing their contribution to the emerging Bitcoin collectibles market and exploring the concept of fungibility within the Web3 ecosystem.

5.1.3. Bridging applications

Bridging applications in the Ordinal ecosystem are pivotal in enhancing the interoperability and utility of the BRC protocols. These applications facilitate the seamless transfer of assets between Bitcoin and other blockchains (e.g., Ethereum) to expand economic activities and user engagement across networks. For instance, MultiBit [109] significantly simplifies the transfer of BRC-20 tokens to EVM networks, allowing users to bridge and mint tokens between Bitcoin and EVM networks effortlessly. OrdBridge [110] enables secure and easy movement of these tokens between Bitcoin’s native chain and Ethereum. This enhances liquidity and opens up diverse exchange possibilities within the Ethereum ecosystem. Lastly, OrdiZK [111] introduces a groundbreaking two-way anonymizer bridge using zk-SNARKs [112] to create an untraceable connection between Ethereum and Bitcoin transactions.

5.1.4. Launchpad applications

Launchpad applications serve as crucial platforms to introduce BRC tokens to the market. These applications support fundraising, token distribution, and initial exposure to potential investors. They emphasize security and community engagement. For example, Turtsat [113] is a community-led platform that facilitates the development of the Ordinal ecosystem. It offers community-driven funding, a protocol to reduce gas costs, indexing services, and an open-source tool platform for developers. OrdMint [114] operates with transparency and fairness. This democratizes the launch process in BRC to promote decentralization and equal opportunity. In contrast, KoiPond [115] provides a strategic platform for investing in early-stage crypto projects with its native KOI token. It emphasizes fair distribution and seamless integration with diverse projects. Collectively, these launchpad applications revolutionize the entry path for new projects into the market. They ensure fair access and community empowerment within the BRC ecosystem.

5.1.5. Stablecoin applications

Stablecoin applications represent a transformative approach to enhancing liquidity, stability, and asset functionality on the Bitcoin network. These applications leverage the capabilities of L2 protocols like the RSK sidechain, which enables smart contracts on the Bitcoin blockchain. This integration unlocks innovative functionalities not readily available on the base layer. One such example is BitStable [121]. This decentralized asset protocol utilizes a dual-token system and a cross-chain structure. This design facilitates the creation, trading, and management of synthetic assets, ultimately improving both liquidity and utility within the Bitcoin ecosystem. In contrast, BitSmiley [120] addresses specific shortcomings in existing stablecoin and lending infrastructure. It allows users to mint bitUSD, a stablecoin pegged to the US dollar, by over-collateralizing their Bitcoin holdings. This approach fosters a more robust and reliable lending environment. Collectively, these applications and services illustrate the potential of BRC protocols to revolutionize the Bitcoin ecosystem. By providing innovative solutions for asset management and stability, they pave the way for a more versatile and robust financial infrastructure.

5.2. Inscription-based services and bots

Inscription-based services refer to platforms and tools designed to facilitate the creation and management of Ordinal, digital artifacts, and NFTs on the Bitcoin network. Some of these services are OrdinalsBot [116], Magic Eden [8], Gamma [117], OrdinalHub [118], and Ordinal Punks [119]. These services offer diverse functionalities such as encoding files into hexadecimal data, facilitating trustless marketplace transactions between Ordinal and Bitcoin, and providing no-code platforms for easy Inscription of various formats like JPEGs and text. They also aim to simplify the technical complexities involved in Ordinal creation, making them accessible to a broader audience. Some services also integrate unique features like “Inscription Gating” for digital communities and the use of Ordinal as collateral for bitcoin-denominated loans. Inscription gating refers to a system where access to digital communities, content, or resources is restricted to those who hold a specific Ordinal on their wallet. It acts as a verification gate, similar to owning a ticket or membership badge. Overall, the Inscription-oriented services are pivotal in enhancing the usability and mainstream adoption of Bitcoin NFTs to showcase their potential as a new form of digital expression and value.

Meanwhile, Inscription Bot is a newer type of automated tool that introduces novel functionalities such as automated token creation, wallet management, and smart contract interactions to the Ordizk blockchain environment [111]. It enables the enhancement of NFTs, tokens, and various digital assets by embedding supplementary data into them. This capability of data embedding introduces an additional layer of functionality to these assets. It facilitates the creation of distinctive token variations, each possessing unique attributes or capabilities. Such an

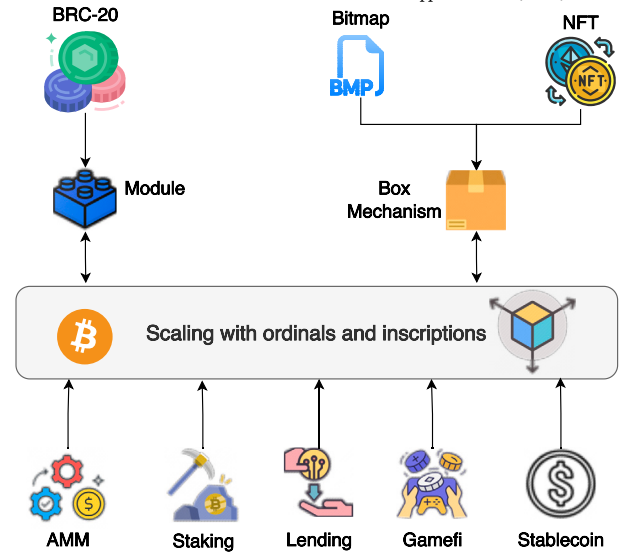


Fig. 8. Ordinals and Inscriptions scaling through BRC-20.

advancement not only broadens the scope of digital assets within the blockchain ecosystem but also paves the way for novel approaches in digital asset management.

5.3. BRC-enabled scaling solution

Scaling diverse applications such as AMM, staking, lending, gamefi, and stablecoins in Bitcoin has been a challenge due to the inherent limitations of the blockchain’s UTXO model. BRC protocols (e.g., BRC-20) offer a novel scaling solution for these applications, particularly within the expanding Metaverse landscape. This approach is inspired by the modular concept proposed by Unisat [105] and Layer 1 foundation [122]. As depicted in Fig. 8, the scaling solution revolves around the integration of BRC protocols with Ordinal and Inscription. The central idea is to introduce a new “Box” mechanism to address the issue of Bitmap and NFTs’ immutability. The Box serves as a container that encapsulates a variety of digital assets, allowing for more sophisticated transaction types and enhanced security measures. In the context of the Metaverse and Web3, the integration of BRC protocols allows seamless management and transfer of virtual assets (e.g., in-game currencies, digital collectibles) within the secure and decentralized environment of Bitcoin. This scaling solution represents a critical advancement in the utilization of Bitcoin for more than just simple transactions, potentially unlocking a new era of innovation in blockchain technology and virtual economies.

6. Research challenges and future directions

The BRC protocol introduces innovative digital asset creation and management capabilities within the Bitcoin network. However, for widespread adoption, it is crucial to address inherent challenges related to Inscription complexity, Ordinal numbering, user-friendliness, existing tool limitations, and long-term stability. This section outlines these key challenges and proposes potential solutions to enhance the BRC protocol’s efficiency and usability.

6.1. Management of UTXO and its complexity

The BRC protocols offer significant flexibility through the use of Inscriptions to encode token data; however, this flexibility introduces considerable challenges. For example, each token operation, such as issuance, transfer, and smart contract creation, involves multiple transactions. This leads to UTXO proliferation, which severely limits Bitcoin’s

efficiency and manageability by contributing to network congestion and increased processing loads. To address these concerns, several innovative approaches are considered. Firstly, adaptive Inscription algorithms can dynamically adjust Inscription processes based on current network conditions, minimizing UTXO generation. These algorithms optimize the process by determining the most efficient way to encode and manage transactions in real-time. Secondly, techniques for more efficient block space utilization streamline transaction processes and reduce data redundancy. By maximizing the use of available block space, it is possible to handle more transactions within a single block, thus minimizing the number of UTXOs created. Thirdly, recursive Inscription methods can allow for more complex token structures without significantly increasing the number of UTXOs. This method involves layering Inscriptions in a way that maintains complexity within a single UTXO, reducing the need for multiple outputs. Additionally, solutions specifically aimed at reducing UTXO generation are critical. This includes sophisticated transaction types that bundle multiple operations into a single transaction. Such approaches can significantly cut down on the number of UTXOs generated per operation. These potential solutions aim to enhance the efficiency and manageability of the Bitcoin network, paving the way for broader adoption and more robust applications of BRC tokens.

6.2. Ordinal numbering and inscription integrity

The current Inscription numbering scheme within BRC protocols presents a vulnerability to data integrity and long-term stability [18]. Potential errors in numbering can compromise the accuracy of the entire transaction history. To address this critical issue, robust mechanisms for error detection and correction are necessary. Furthermore, the use of diverse Inscription methodologies across different BRC iterations introduces additional complexity and potential for errors. Notably, ensuring consistent numbering across distinct chains and mitigating the risk of Inscription collisions or conflicts are critical considerations. Therefore, exploring alternative numbering schemes and refining error-correction mechanisms are crucial for enhancing data integrity and upholding a stable and reliable numbering system.

6.3. Limited compatibility with smart contract

The BRC ecosystem currently lacks a comprehensive suite of tools and supporting services compared to established token standards such as ERC-20 on Ethereum. This limitation makes it challenging for users and developers to interact effectively with BRC tokens. Bitcoin's design inherently prioritizes security and simplicity over complex programmability. In contrast to Ethereum, which is built to support a wide array of decentralized applications through advanced smart contracts, Bitcoin's scripting language is more limited. This restriction affects Bitcoin's capacity to handle complex transactions and scalable applications, especially those that require sophisticated logic and interactions, such as DeFi protocols [12]. Therefore, it is essential to build robust infrastructure, including exchanges, wallets, and development tools, to empower users and developers to leverage the full potential of BRC tokens. Meanwhile, it is important to ensure that platforms and tools can manage the unique features of newer and experimental BRC tokens to enhance their usability and adoption. Integrating these measures would improve infrastructure and enhance the liquidity of BRC tokens, promoting widespread adoption and utilization of DeFi and Web3.

6.4. Security concerns

The National Vulnerability Database (NVD) of the United States has identified a significant cybersecurity vulnerability within Bitcoin's Ordinals protocol [123]. This vulnerability allows for bypassing the data carrier limit by disguising data as code in certain versions of Bitcoin Core [124] and Bitcoin Knots [125]. As a result, non-transactional data

can be added to the blockchain. With the increasing popularity of Bitcoin Inscriptions, this ecosystem has become a target for various scams. For instance, scammers create JSON fields for transferring Inscriptions and encode them as hex for users to inscribe. This can potentially lead to the theft of users' Inscriptions [126]. Additionally, on trading platforms, the presence of numerous Inscriptions with identical names makes it difficult to identify authentic ones. Scammers exploit this by creating fake Inscription series and adding invalid fields to mimic genuine ones. Addressing these vulnerabilities and developing robust verification mechanisms are critical to enhancing the security and trustworthiness of the Bitcoin ecosystem.

6.5. User experience and accessibility

Compared to established standards like ERC-20 on Ethereum, BRC tokens currently lack user-friendly interfaces and tools for token creation, deployment, minting, and transfer. This limitation hinders user and developer adoption and affects the ecosystem's growth. The complex Inscription process and limited functionality of BRC protocols, when compared to smart contract-enabled platforms, create significant barriers in terms of accessibility and usability. Additionally, the absence of intuitive platforms, tools, and support services makes it challenging for users and developers to effectively interact with BRC tokens. Therefore, enhancing the user experience through simplified processes, comprehensive documentation, and robust application packages is crucial for broadening adoption and fostering a dynamic DeFi and Web3 ecosystem.

6.6. Interoperability and liquidity issues

Inscription protocols face significant challenges in achieving interoperability and ensuring robust liquidity, both critical for the expansion of the BRC ecosystem. Developing infrastructure to enhance liquidity through interoperable mechanisms remains essential [127]. In 2023, Ordinals Market and Bitcoin Miladys collaborated to introduce the BRC-721E token standard [128], enabling the transfer of ERC-721 NFTs from Ethereum to Bitcoin Ordinals via burn transactions. These transactions encode NFT data and specify a Bitcoin address for Inscription receipt, but their irreversible nature prevents converting Ordinals back to ERC-721 NFTs [80], posing a key research challenge. To address interoperability, several cross-chain bridging protocols have emerged to enable seamless asset transfers across blockchain networks. For example, MultiBit [109] supports BRC-20 to ERC-20 conversions, Teleport-DAO [129] enables trading between Bitcoin and Polygon, and SoBit Bridge [130] connects BRC-20 assets to Solana by minting equivalent tokens. Additionally, decentralized exchanges, such as Allins [131] and Ordswap [132] leverage AMM models to enhance Inscription liquidity, allowing users to trade and earn revenue through smart contract-based pools.

Beyond these initial developments, a new wave of platforms is emerging to enable more advanced and direct interoperability across Inscription ecosystems. For example, Space-Zero [133], an omnichain bridge, facilitates secure transfers of Inscription assets across Ethereum, BNB Smart Chain, and various L2 networks. Similarly, SatoshiSync [134] integrates with Portal DEX to enable bridgeless atomic swaps, thereby enhancing liquidity without dependence on native assets. Likewise, UniRouter [135] connects Bitcoin Inscriptions to the EVM ecosystem, enabling seamless interaction with smart contracts and DeFi platforms. Furthermore, InsLink [136] provides a plugin-based bridge architecture that supports multi-chain Inscription mobility with minimal integration overhead. On DogeChain, OxDogeSwap [137] facilitates trading across distinct Inscription protocols, contributing to cross-ecosystem composability. Despite these innovations, key technical and architectural challenges persist [138]. The reliance on centralized indexers introduces potential vulnerabilities, including inconsistent state tracking and data integrity issues. Fragmentation among competing standards leads to

asset silos and diminished interoperability. Consequently, secure cross-chain execution remains an open problem, necessitating robust solutions such as multi-signature verification and trust-minimized communication. Collectively, these advancements and ongoing challenges highlight the critical need for unified, scalable, and secure infrastructure to support the evolving Inscription ecosystem.

6.7. Scalability challenges and L2 solution

Bitcoin's base layer faces significant scalability challenges, primarily due to its limited transaction throughput, high fees during periods of peak demand, and inefficient on-chain data storage. The network processes approximately 7 transactions per second (TPS), with a block size capped at 1 MB every 10 min, leading to congestion and fee spikes when transaction demand surges [139]. The introduction of Ordinals and BRC-20 tokens exacerbated these issues by embedding additional data into Bitcoin blocks, as both require multiple on-chain transactions for actions such as minting and transferring. For instance, BRC-20 transactions often require several Inscriptions to split token amounts due to fixed increment constraints, significantly increasing transaction volume and clogging the Bitcoin mempool [140]. This inefficiency results in substantial fee increases, as users compete in priority gas auctions to process token-related activities. Moreover, the permanent storage of Inscription data, ranging from text and images to video games, on the blockchain has led to significant data bloat. By September 2023, over 33 million Inscriptions had been created, expanding the UTXO set and raising concerns about the growing costs of running full nodes [138]. These challenges highlight the urgent need for scalable solutions to support the increasing use of Bitcoin-based tokens and Inscriptions.

To address these scalability limitations, Bitcoin L2 solutions such as RSK, Stacks, and LN offer specific mechanisms for integrating with BRC tokens and Ordinals while mitigating throughput, fee, and storage constraints. For example, RSK is a sidechain secured through merged mining with Bitcoin. It features an Ordinal-aware bridging mechanism named RSKIP-387 to facilitate the transfer of Ordinals between Bitcoin and RSK [141]. This mechanism locks inscribed sats in the RSK peg and mints corresponding NFTs on RSK for efficient trading, allowing users to redeem the original Ordinals on Bitcoin. For BRC-20 tokens, RSK's support for RRC-20 (ERC-20-like tokens) enables equivalent token creation, although a direct bridge is still under development. On the other hand, Stacks leverages Proof of Transfer (PoX) consensus and the sBTC peg. It introduces the Stacks Ordinals, which wraps Ordinals as NFTs on the Stacks chain to enable faster and cheaper transactions while maintaining a connection to Bitcoin's security [142]. Additionally, Stacks supports native token standards, such as SIP-010 to provide a more efficient alternative for hosting BRC-20-like tokens. The Lightning Network, designed for high-volume micropayments, indirectly reduces network congestion by offloading BTC transactions. For instance, the OmniBOLT integrates BRC-20 tokens into the LN to allow off-chain transfer of BRC tokens [143]. Looking ahead, these L2 solutions are progressing toward deeper integration with BRC protocols. As these technologies mature, they will play an essential role in supporting the growth of the BRC token ecosystem, enabling more efficient token transfers, reducing on-chain congestion, and paving the way for a more scalable and flexible Bitcoin network.

6.8. Cross-chain security and compatibility

While the development of cross-chain platforms enhances the reach of BRC tokens, Ordinals, and Inscriptions, their practical operation introduces specific security and compatibility challenges. These include vulnerabilities in token transfer processes and concerns over the operational integrity of third-party bridges or custodians, along with compatibility issues arising from differences in blockchain architectures. Recent works [144,145] have identified Inscription-specific exploits,

such as liquidity-freezing pinning attacks and double-spend replay vulnerabilities. More broadly, the bridge mechanisms vital for cross-chain transfers remain frequent attack vectors, exhibiting flaws such as replay attacks that generate unauthorized tokens, authentication bypasses, and incorrect deposit or proof-of-burn logic, leading to substantial financial losses [146,147]. These security risks are further exacerbated by the common reliance on centralized indexers for state tracking. At the same time, interoperability between chains introduces several technical challenges. These include mismatches in state-mapping models and inconsistencies in decimal precision when bridging between differing ledger systems (e.g., Bitcoin's UTXO model versus Ethereum's account-based model) or incompatible token standards [148]. Moreover, unique Ordinal attributes, such as "cursed" Inscriptions can create ambiguities in resolution and provenance on non-native chains [149]. The lack of standardization among competing Inscription protocols further limits cross-chain operability. As a result, achieving secure and reliable cross-chain execution remains an unresolved problem. Addressing these issues requires robust technical solutions, including harmonized metadata schemas, standardized numeric conventions, advanced cryptographic verification (e.g., multi-signature schemes), trust-minimized communication protocols, decentralized validation frameworks, and formally verified smart contracts for bridges. These measures are essential to support a unified, scalable, and secure Bitcoin-based ecosystem.

6.9. Long-term stability

Initial concerns about the long-term stability of the Inscription numbering scheme have raised concerns regarding the future robustness of BRC protocols. As BRC tokens evolve and undergo successive iterations, it is important to maintain their stability and reliability. This maintenance is crucial to effectively address increasingly complex challenges. Moreover, sustained research and development efforts are essential. These efforts aim to continually refine and enhance these protocols in response to emerging issues. Additionally, fostering greater engagement within the broader BRC community is crucial. This engagement is necessary for soliciting feedback and promoting collaborative solutions, which are vital for sustainable development. By nurturing a cooperative ecosystem, BRC protocols can remain flexible and responsive as they need to adapt to the dynamic needs of the Bitcoin space and its ongoing technological advancements. Such adaptability is essential to ensure the long-term viability of BRC tokens and their integration within the broader blockchain infrastructure.

6.10. Impact on real-world asset tokenization

The tokenization of RWAs through BRC protocols introduces unique research challenges. For instance, the RWA-20 protocol [150] faces the fundamental challenge of bridging on-chain inscriptions with off-chain asset states. This requires efficient oracle mechanisms and verifiable proof-of-ownership systems to ensure legal compliance across multiple jurisdictions [151]. Unlike traditional tokenization approaches, RWA inscriptions must encode complex metadata including legal documentation, compliance certificates, and asset-specific attributes within Bitcoin's constrained data limits [152]. Future research must address the reversibility challenges where immutable blockchain records must accommodate real-world events such as asset destruction, legal disputes, or ownership reversals through court orders. Additionally, the development of decentralized custody solutions for physical assets, standardized valuation methodologies for on-chain price discovery, and cross-jurisdictional settlement mechanisms are the critical areas that require interdisciplinary collaboration between researchers, legal scholars, and traditional finance institutions.

6.11. Regulatory challenges and potential solutions

The integration of BRC protocols, Ordinals, and Inscriptions into the financial ecosystem necessitates careful scrutiny of their regulatory

standing due to inherent legal ambiguities. A primary hurdle lies in the classification of these assets, which may qualify as securities, utility tokens, commodities, or digital collectibles. For instance, the U.S. *Howey* Test can classify BRC tokens as securities if they are marketed with an expectation of speculative profit [153]. Likewise, Ordinals can be deemed securities if sold as investment contracts, as highlighted in the U.S. Securities and Exchange Commission *Impact Theory* case [154]. This regulatory uncertainty is global. For example, Switzerland's FINMA has introduced utility token classifications [155], while the EU's Markets in Crypto-Assets (MiCA) regulation governs issuers and Crypto-Asset Service Providers [156]. Other jurisdictions, including the UK's Financial Conduct Authority, Hong Kong's Securities and Futures Commission, and Singapore's Monetary Authority of Singapore are also developing regulatory frameworks. The diversity of global approaches leads to regulatory arbitrage, enforcement challenges caused by decentralization, and difficulties in balancing innovation with investor protection. Bitcoin's technical constraints, combined with taxation complexities such as asset valuation, transaction tracking, and recognition of NFTs as collectibles [157] further complicate the regulatory landscape.

To address regulatory challenges, it is important to ensure the clearer global taxonomies, adaptive legal frameworks, and stronger international cooperation. Potential solutions include precise legal definitions for crypto-assets, drawing from FINMA [155] and MiCA [156]. Additionally, adaptive securities regulations, such as safe harbor provisions or targeted legislation for digital commodities, can promote innovation while maintaining safeguards. Tailored NFT regulations may involve marketplace licensing and enhanced consumer protection mechanisms. Strengthening Anti-Money Laundering (AML) and Counter-Financing of Terrorism frameworks is equally critical. This can be achieved through improved transaction monitoring and broader application of compliance mandates, such as those outlined in the Financial Action Task Force Travel Rule and its national implementations (e.g., in South Africa) [158]. Global institutions, such as the Financial Stability Board and the International Organization of Securities Commissions, advocate for regulatory harmonization based on the principle of "same activity, same risk, same regulation" [159]. Finally, industry self-regulation, RegTech-driven compliance solutions, and robust investor education will be essential for enabling responsible innovation and supporting the long-term stability and mainstream adoption of BRC protocols and Ordinals.

6.12. Alternative protocols

The increasing interest in Inscriptions has prompted the adoption of similar practices across various public blockchains, leading to a variety of derivative protocols and imitation projects. Despite its innovation, the BRC-20 protocol faces issues such as a four-letter token limit and vulnerability to front-running attacks [160]. In response, the ARC-20 protocol [99] was developed to remove the character restriction, thereby broadening the potential for more varied applications. Within this expanded framework, Realm [161] stands out by allowing each registered prefix text to control pricing rights over all corresponding suffixes. Similarly, Rune [89] facilitates the issuance of fungible tokens by embedding token data within UTXO scripts. Unlike BRC-20, which relies on external data storage and inefficient multi-step transactions, Runes leverages Bitcoin's native UTXO model and OP_RETURN data for a streamlined and index-friendly solution [162]. This design seamlessly integrates with platforms such as LN [43]. Another significant development is RGB [163], which enhances scalability by converting smart contract states into concise proofs inscribed within Bitcoin UTXO output scripts. This approach offers low transaction costs and high scalability, effectively serving as an L2 solution while leveraging Bitcoin's inherent security.

Furthermore, DRC-20 [164] mirrors BRC-20 but is particularly popular because of its lower transaction costs and significant cultural appeal. Consequently, the Etherscriptions protocol [165] introduces "dumb

contracts" that enhance the functionality and utility of Ethereum. Protocols like ASC-20 [166], BSC-20 [167], and PRC-20 [168] on other EVM-compatible chains facilitate Inscriptions and indexing for notable projects (e.g., Polkastarter [169]). On the Solana blockchain, SOLs [170] focus primarily on NFTs and utilize image or file hash-based indexing. The SFNs [171] represent the initial Inscription effort on the EOS chain and adopt the AtomicAsset standard [172] to optimize feature integration while prioritizing RAM efficiency. Moreover, the MRC-20 [173] on the Move blockchain and APT-20 [174] on the Aptos blockchain are experimental token standards that leverage the Ordinals protocol to streamline the creation and transfer of fungible tokens.

7. Conclusions

This paper presents a comprehensive examination of the Ordinals and Inscriptions protocols, underscoring their transformative potential in redefining Bitcoin's utility within the broader Web3 landscape. By enabling the direct embedding of structured data into Bitcoin transactions, these protocols extend the blockchain's functionality beyond simple transfers, supporting the emergence of novel asset classes and decentralized applications. We analyzed the core technical mechanisms underpinning Ordinals and Inscriptions, along with their practical applications, with a particular focus on their implications for cross-chain interoperability and ecosystem integration. Furthermore, we surveyed the evolution of BRC token standards and assessed their role in enabling tokenization and decentralized finance on Bitcoin. Despite the rapid innovation, widespread adoption is limited by persistent challenges related to scalability, interoperability, and usability. Our study identifies these critical barriers and outlines potential pathways for advancing protocol efficiency, security, and accessibility.

In the future, we aim to emphasize the integration of AI and ML to advance Ordinals, Inscriptions, and BRC protocol implementation and interoperability. Specifically, we will investigate how AI-driven solutions can support the development of dApps to enable more seamless cross-chain interactions [45]. One promising direction involves exploring AI-native protocols such as BRC-720 [175], which enables the automated creation of dynamic digital assets, such as interactive NFTs inscribed on individual sats. This approach can also be extended to support the generation and customization of BRC-20 token metadata. For instance, the HUMAN AI project [176] leverages generative AI to create high-resolution art collections inscribed on Bitcoin Ordinals, demonstrating the potential for AI to automate and enrich digital asset creation within the BRC ecosystem. Likewise, the combination of GenAI and large language models holds considerable potential. For instance, ChainGPT's Ordify [177] leverages AI analytics to optimize BRC-20 token launches by applying LLMs to forecast market dynamics and enhance investor engagement during initial DEX offerings. Moreover, the role of AI agents in Bitcoin protocols will be further analyzed. For example, Xverse [178] employs AI agents to automate wallet interactions, validate BRC-20 transfers, and improve both user accessibility and operational security. Similarly, the Goose framework [179] supports AI agents that interact with Ordinals Inscriptions via the Model Context Protocol, enhancing automation and user experience in managing Inscription-based assets. Another emerging project, Sophon [180], utilizes AI to optimize transaction batching in zero-knowledge rollups, thereby addressing scalability challenges by processing Inscriptions off-chain while maintaining data integrity. Additionally, AI-driven content management solutions, such as the YOUR AI protocol [181] use BRC-20 Inscriptions to enable personalized content streaming and monetization, showcasing AI's role in optimizing dApp functionality. Collectively, these directions represent a multifaceted research agenda aimed at advancing the BRC, Ordinals and Inscription ecosystem through intelligent automation, improved security, and enhanced interoperability for accelerating Web3 adoption.

Funding

This work is supported by the National Key Research and Development Program of China (No. 2020YFA0909100), and the Basic and Applied Basic Research Foundation of Guangdong Province (No. 2023TQ07A264).

CRediT authorship contribution statement

Md Monjurul Karim: Writing – original draft, Conceptualization, Methodology. **Qiang Qu:** Validation, Methodology, Funding acquisition, Writing – review & editing, Supervision, Investigation, Conceptualization. **Yaoyi Cai:** Conceptualization, Writing – review & editing. **Teddy Liu:** Writing – review & editing. **Xingchen Meng:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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